HISTORY OF HCI-AERO TOWARD HUMAN SYSTEMS INTEGRATION

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AIR & SPACE ACADEMY
INTERNATIONAL ACADEMY OF ASTRONAUTICS
INCOSE FELLOW

MY WORLD FOR THE LAST ~45 YEARS …

… in technical design
and operations

From correction to interaction to integration

Human Factors & Ergonomics …
Human-Centered Design …
Human Systems Integration …

FlexTech

CentraleSupélec-ESTIA Chair

HUMAN SYSTEMS INTEGRATION (HSI) FOR INCREASINGLY AUTONOMOUS SYSTEMS

Digital engineering: human-centered design, flexibility, uncertainty, tangibility, maturity, etc.

Roles of people & organizations in life-critical systems: HSI sociotechnical, ethical & epistemological issues

Development of new approaches, methods & tools, e.g., digital twins, human-in-the-loop simulations, etc.

Applications in various industrial sectors, e.g., aerospace, defense, energy, health, automotive, nuclear, etc.

Design for Flexibility, HSI & Autonomy…

EVOLUTION FROM AUTOMATION ISSUES TO TANGIBILITY ISSUES

The international conference '**HCI-Aero**': **between 1986 and 2016**.

For aeronautical researchers and engineers: **human-machine interaction and AI**.

It gave rise to the field of **Human Systems Integration (HSI).**

INCOSE HSI conferences launched in 2019 (Biarritz, France).

… *I will explain how HSI came about!*

HSI foundations and current prospects, using examples from aeronautics.

Systems engineering, human factors and ergonomics, and information technology…

Numerous examples of applications will be presented.

A FEW DATES…

1986: first conference at ISAE-SUPAERO

1986-1995: HMI-AI-AS conference series

HCI-AERO CONFERENCES
A FEW DATES AND LOCATIONS…

30 years

1986: first conference at ISAE-SUPAERO

1986-1995: HMI-AI-AS conferences

1998: first HCI-Aero in Montreal, Canada

2000: Toulouse

2002: MIT, Cambridge

2004: Toulouse

2006: Seattle

2010: Cape Canaveral, Florida

2012: EUROCONTROL, Brussels, Belgium

2014: NASA Ames & Santa Clara, California

2016: DGAC, Paris

HSI CONFERENCES

- 2019: FIRST HSI CONFERENCE IN BIARRITZ, FRANCE
- 2021: SAN DIEGO, CALIFORNIA, USA
- 2024: JEJU, KOREA
- 2027: DUBLIN, IRELAND

INCOSE
International Council on Systems Engineering

HCI-AERO CONTENT…

Analysis, design and evaluation of…

… cockpit and ATC systems

A forum for industry and academia…

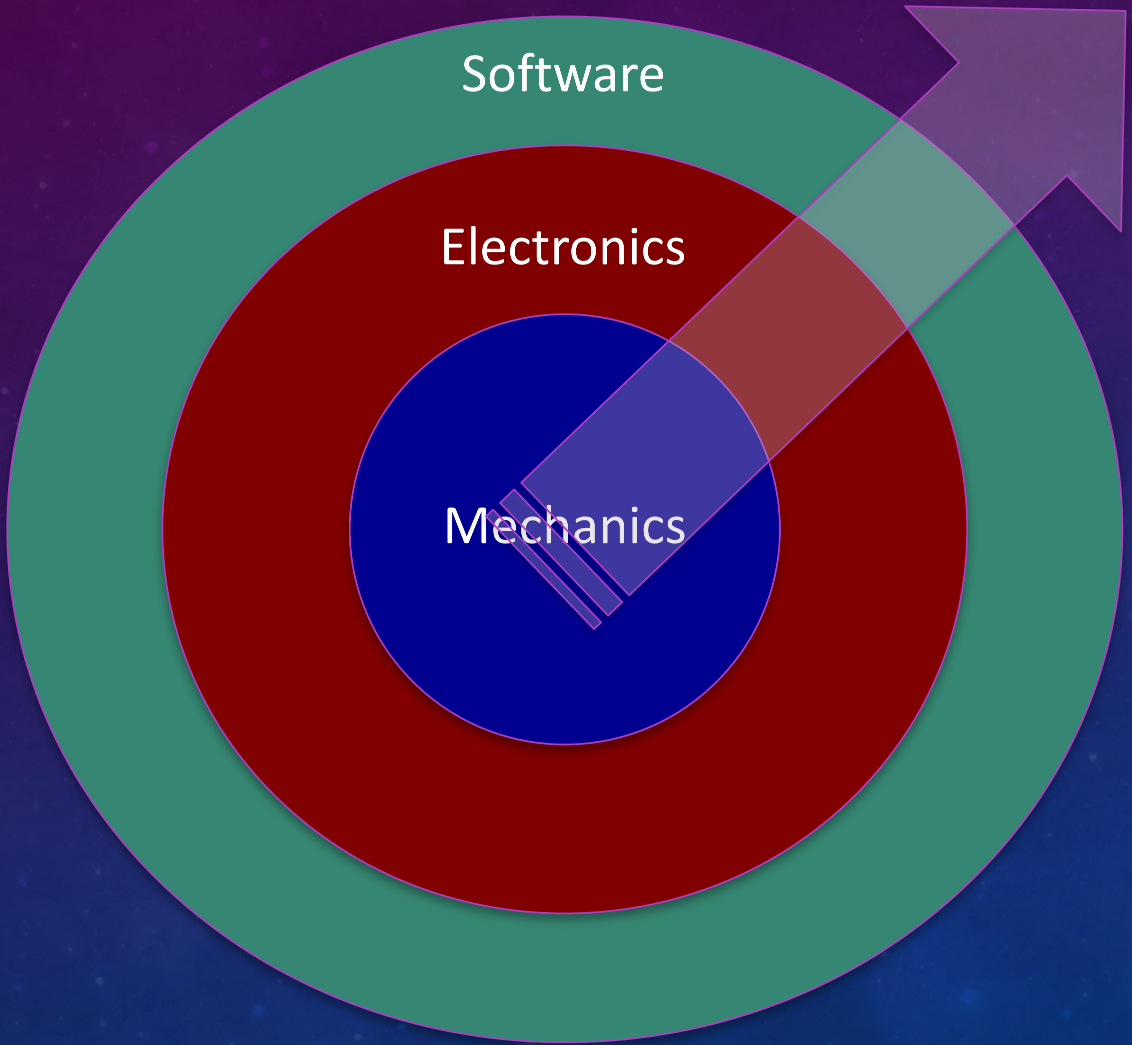
Quality research papers → ACM Digital Library

Latest aerospace technology and practices…

A community!

From Hardware to Software

20th century

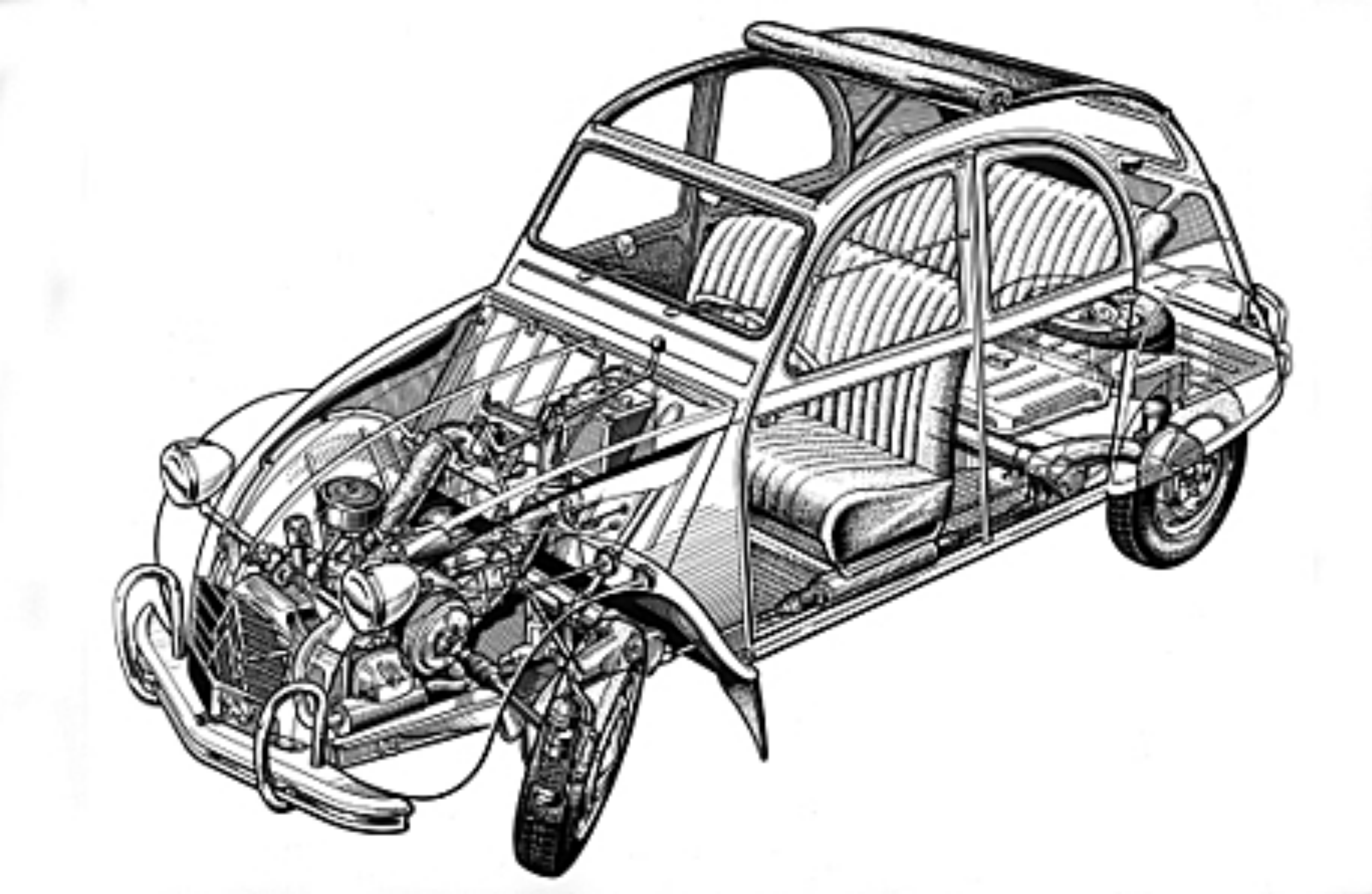


Incremental Accumulation of Artificial Functions into Structure

…

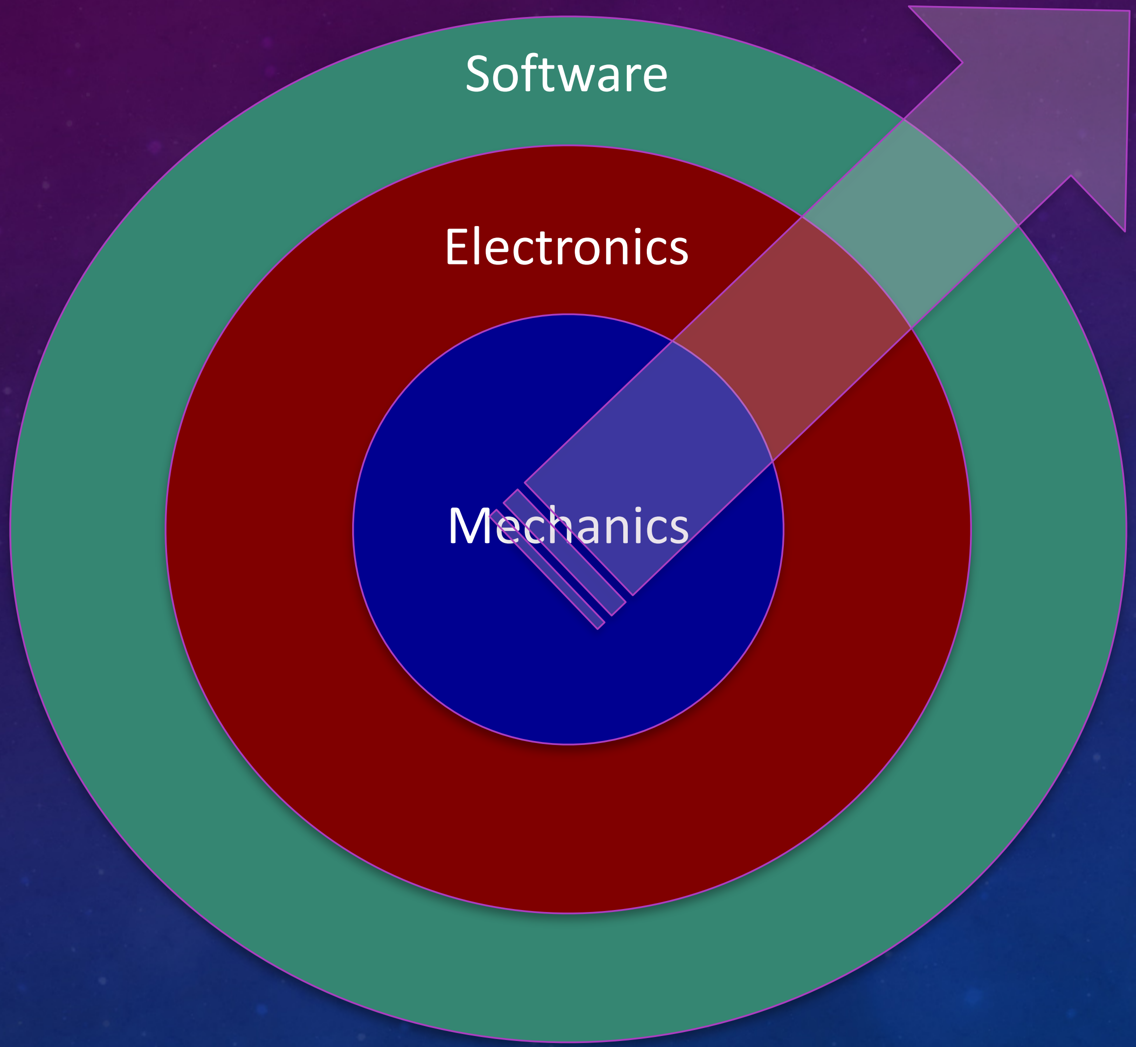
Automation & HCI


714



20th century

From Hardware to Software

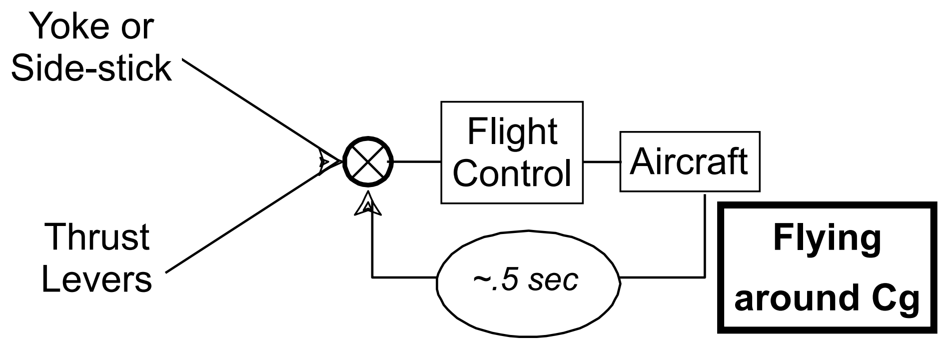


Incremental Accumulation of Artificial Functions into Structure

…

Automation
A.I. & HCI

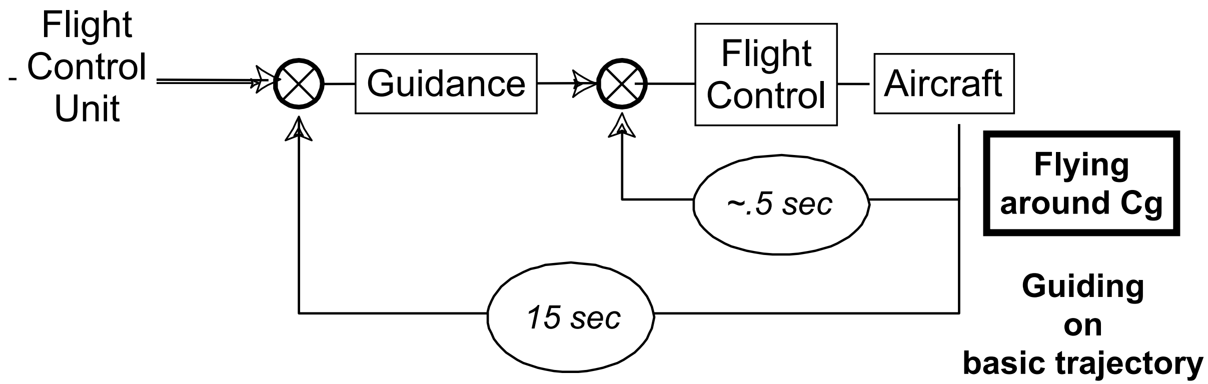
Loop 1: Trajectory control automation



- Single agent
- One parameter at a time

The 4-loop approach to ATM was first provided by Etienne Tarnowski at HCI-Aero'06, Seattle, USA

Loop 2: Guidance automation

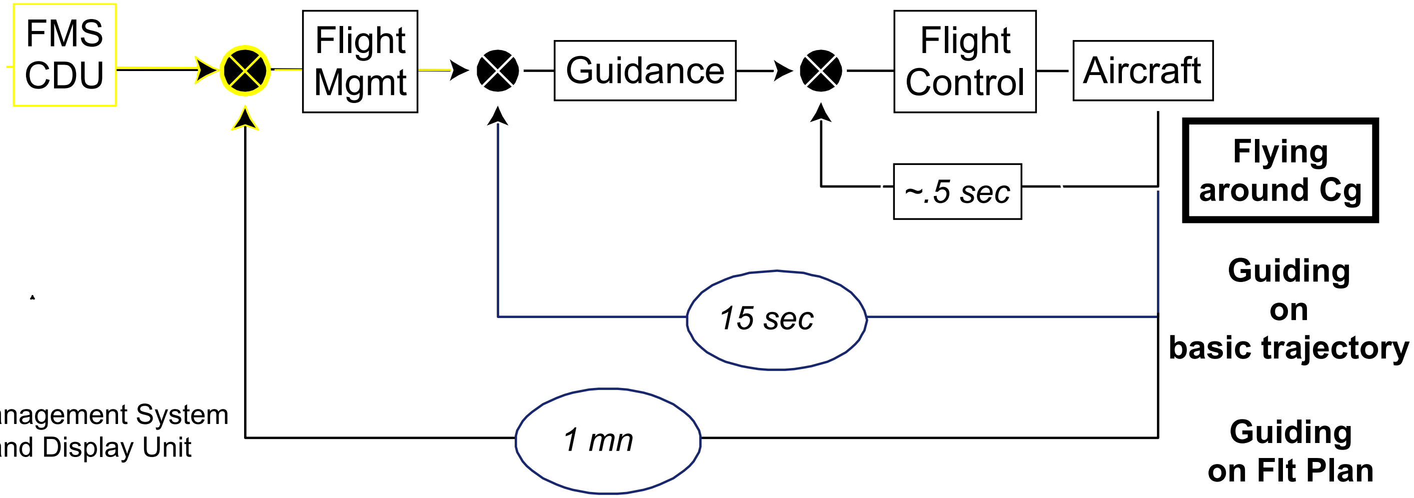


Integrated and digital autopilot and autothrottle

High level modes (evolution)

Loop 3: Navigation automation

Glass cockpits



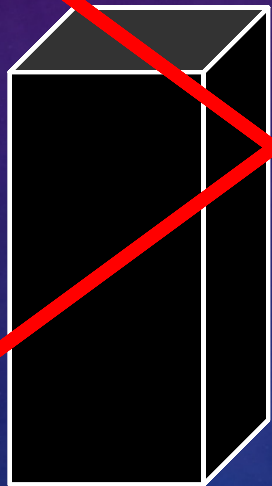
FMS: Flight Management System
CDU: Control and Display Unit

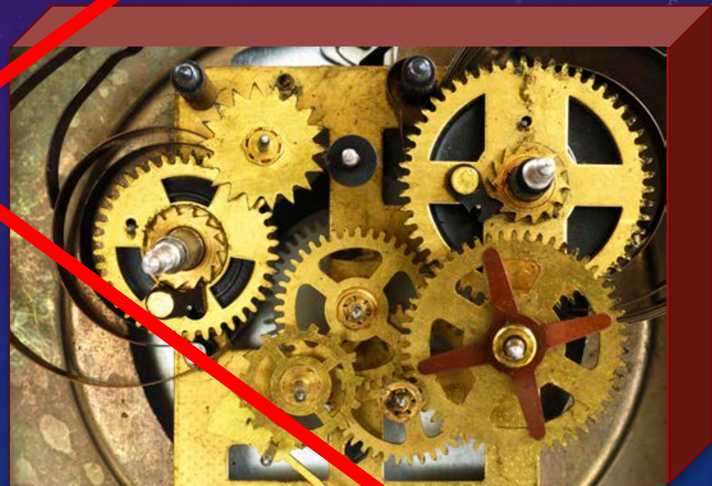
Integration of guidance and flight management

From control to management (revolution)

Historical development of HFE?

User Interface

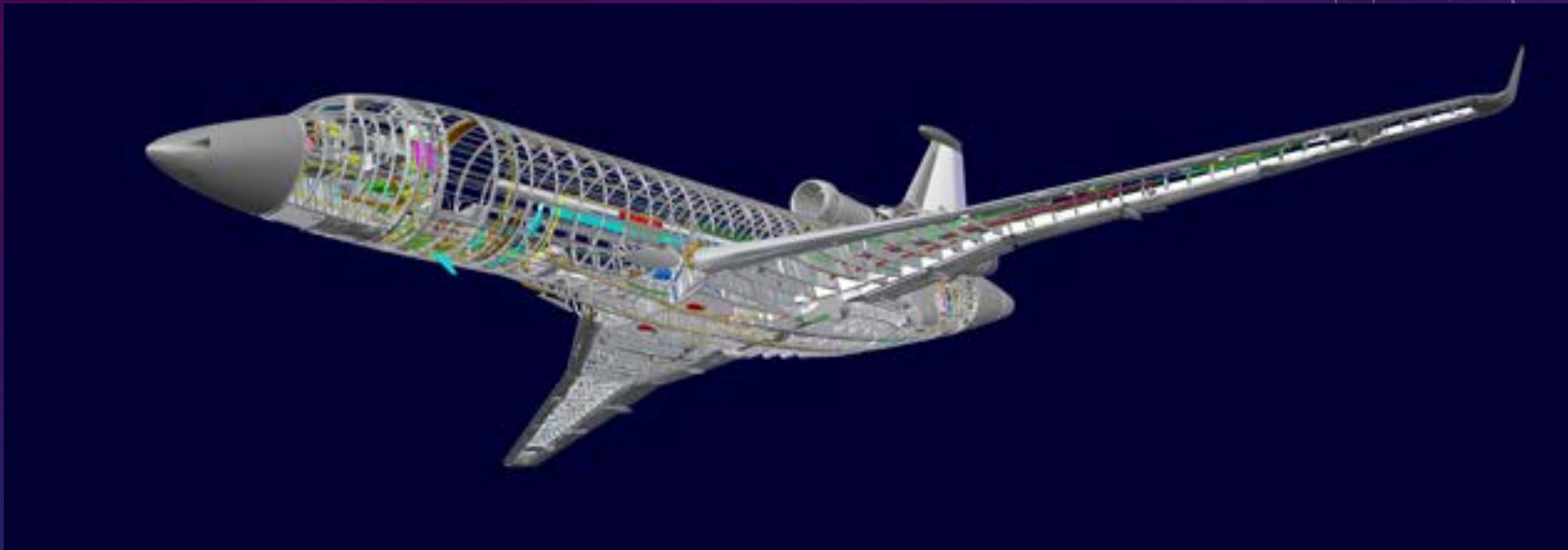




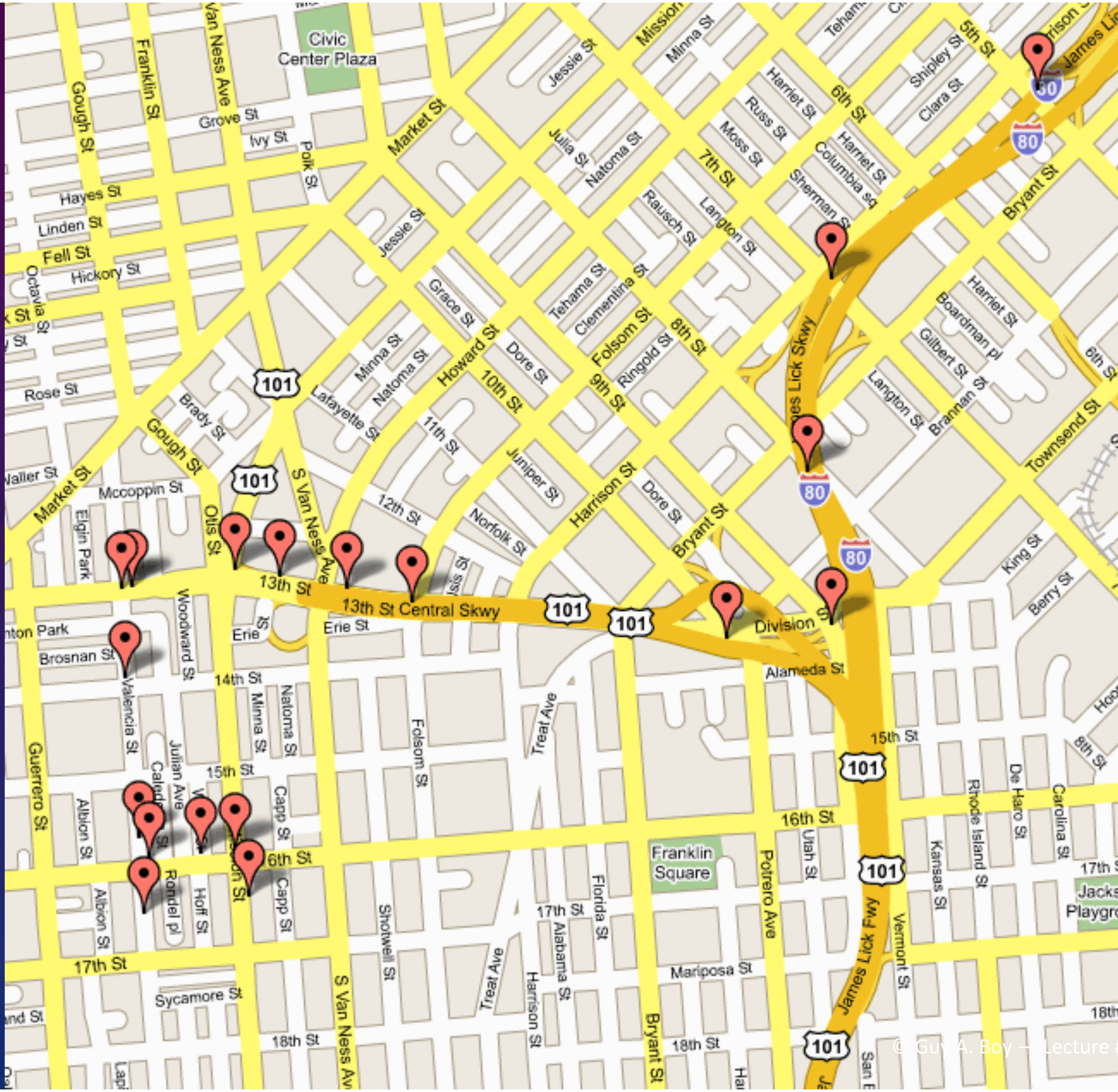
Human Factors

Ergonomics

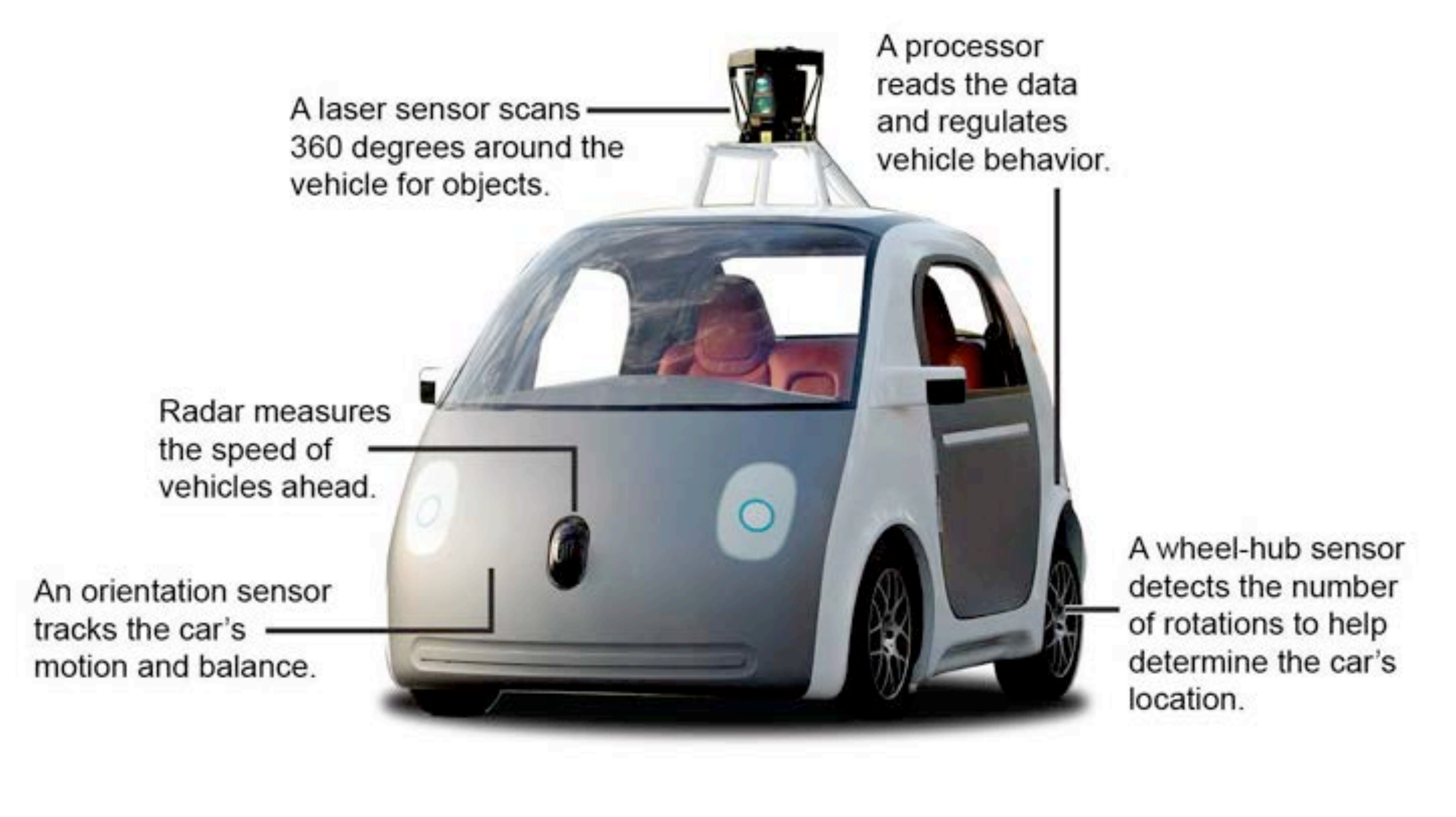
Engineering



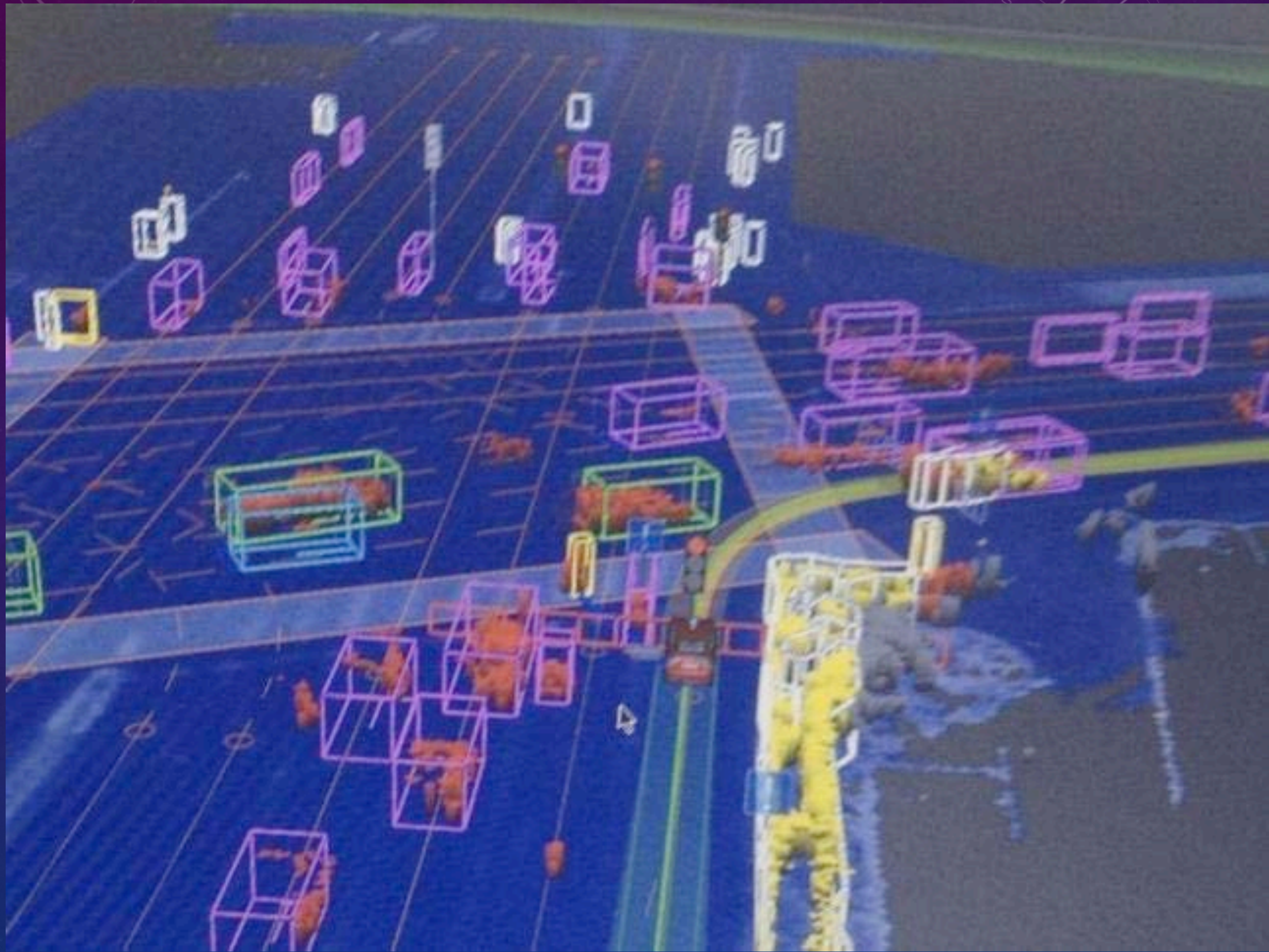
FLIGHT TESTS BEFORE THE AIRCRAFT IS BUILT


Civic Center Plaza
Franklin St
Van Ness Ave
Gough St
Grove St
Ivy St
Polk St
Market St
Hayes St
Linden St
Fell St
Hickory St
Octavia St
Rose St
Jessie St
Mission
Minna St
Julia St
Natoma St
7th St
6th St
5th St
Harriet St
Russ St
Moss St
Columbia sq
Sherman St
Rausch St
Langton St
Shipley St
Clara St
Bryant St
James Lick Fwy
Grace St
Tehama St
Clementina St
Folsom St
Ringold St
8th St
9th St
10th St
11th St
12th St
Howard St
Dore St
Lafayette St
Brady St
Boardman pl
Gilbert St
Brannan St
Townsend St
Juniper St
Norfolk St
Harrison St
Waller St
Mccoppin St
Otis St
S Van Ness Ave
Elgin Park
13th St
13th St Central Skwy
101
80
Division St
Alameda St
Erie St
Brosnan St
Woodward St
14th St
Minna St
Natoma St
Valencia St
Julian Ave
15th St
Capp St
Folsom St
Treat Ave
16th St
Franklin Square
Potrero Ave
Utah St
Vermont St
Kansas St
Rhode Island St
De Haro St
Carolina St
King St
Berry St
Guerrero St
Albion St
Rondel pl
Hoff St
Shotwell St
17th St
Alabama St
Florida St
Mariposa St
Sycamore St
18th St
Harrison St
Bryant St

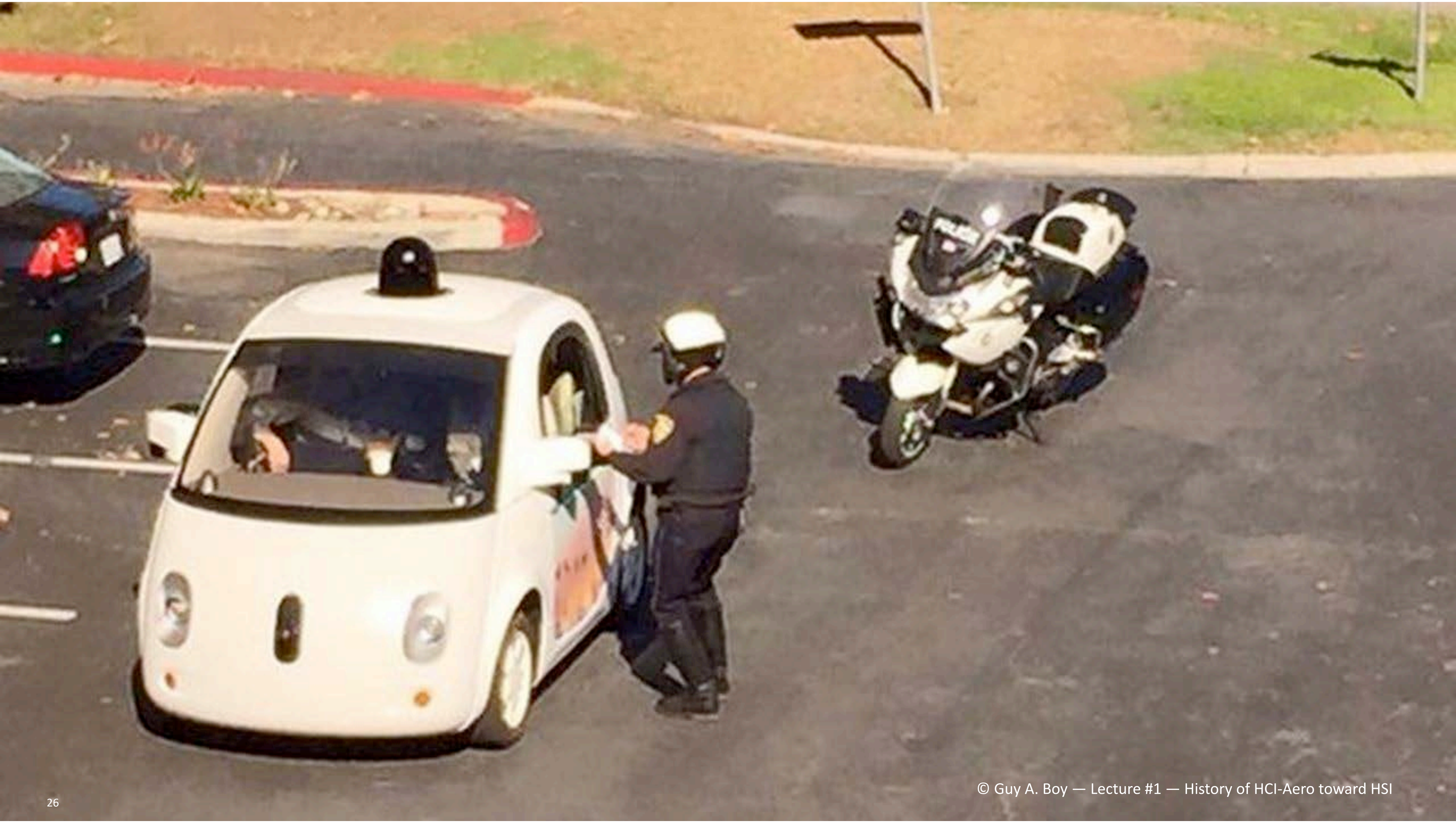
GOOGLE SELF DRIVING CAR



CAR’S MENTAL MODEL…



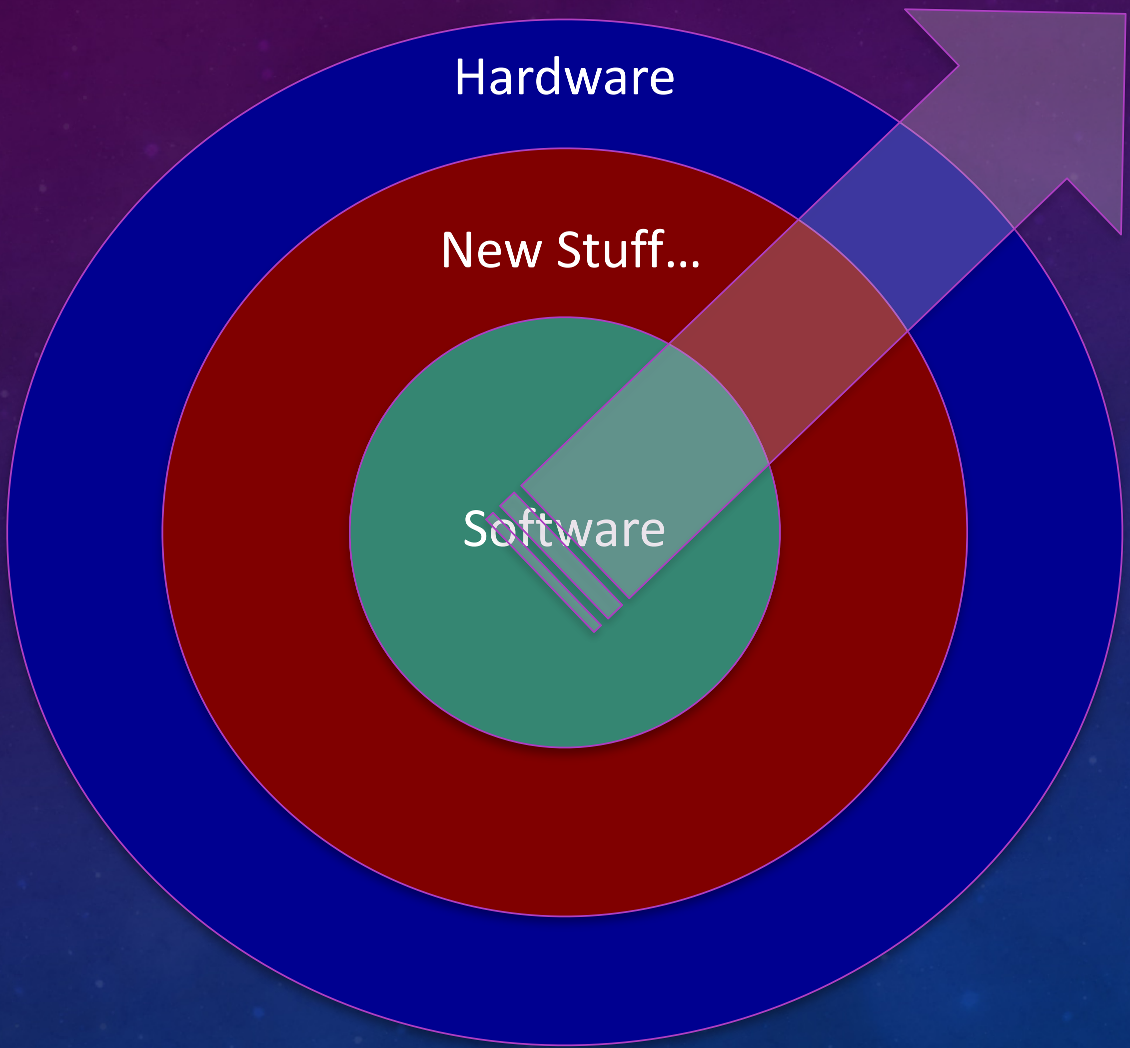
TESTING FOR TANGIBILITY…



POLICE

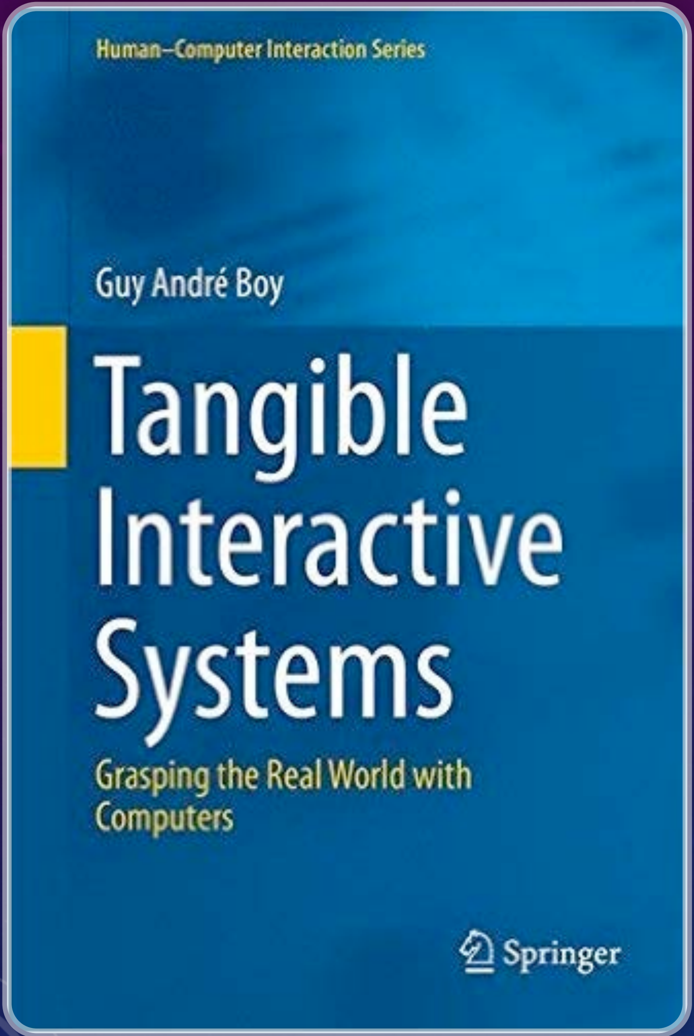
21st century

From Software to Hardware



Modeling
Simulation
Connectivity
Orchestration
3D Printing
…

Tangible Interactive Systems (TISs)



TANGIBLE: WHAT DO WE MEAN?

Something is tangible when it is **graspable**

in the **physical** sense,

but also in the **figurative** sense.

DISCIPLINARY EVOLUTION

Human Factors and Ergonomics (Human-Machine Interfaces)

→ HFE experts correct engineering productions

Human-Computer Interaction

→ From corrective ergonomics to interaction design

Human-Systems Integration

→ Systems engineering and HCD combined

HUMAN-CENTERED DESIGN (HCD)

HCD for whom?

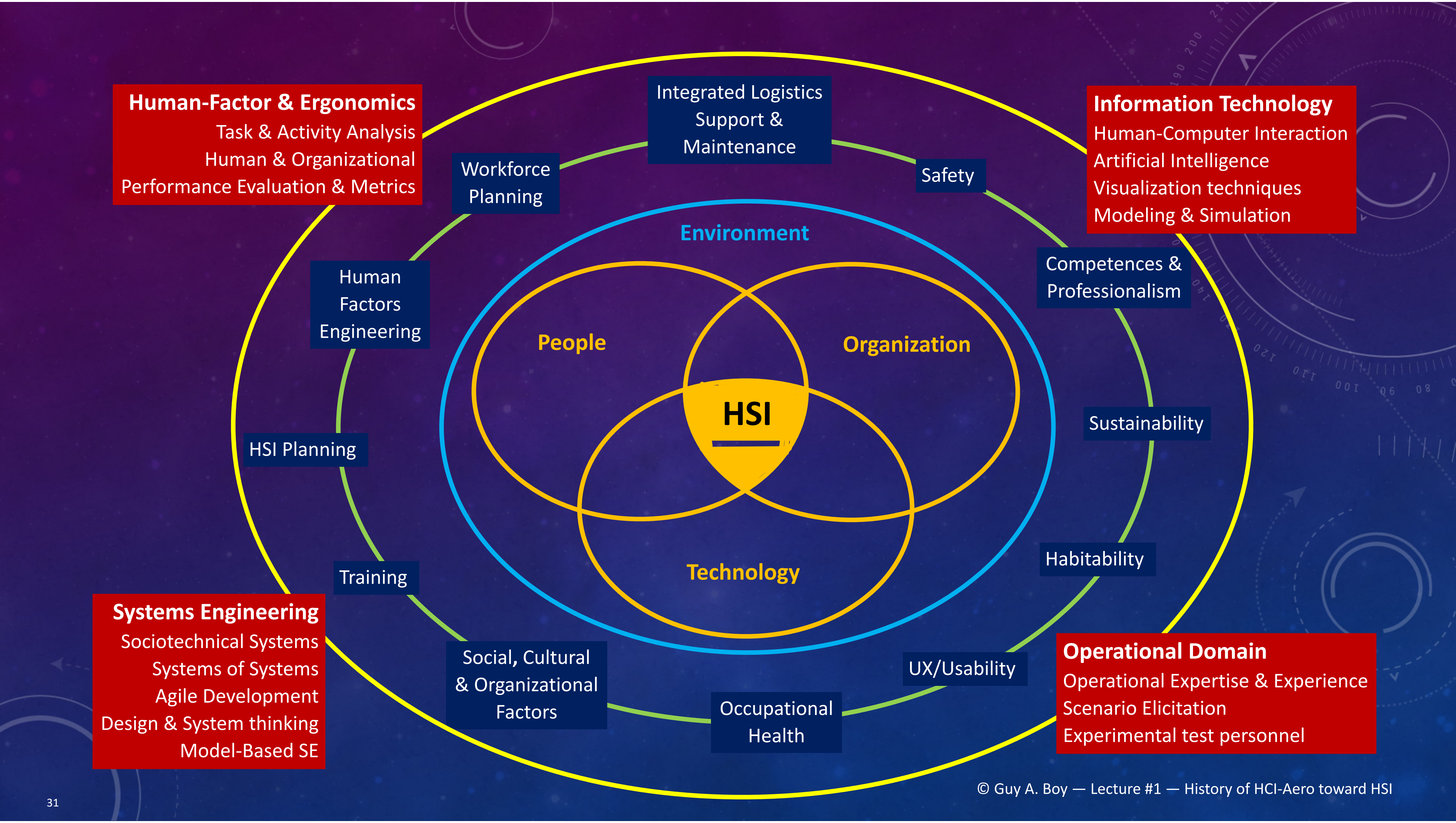
- Pilots, controllers, maintenance personnel, airlines, etc.
- Engineering designers, developers, manufacturers, certifiers, etc.

HCD assumes that there is always the **human element** everywhere

- How do we consider the human element?
- What are the theoretical and practical methods and tools?

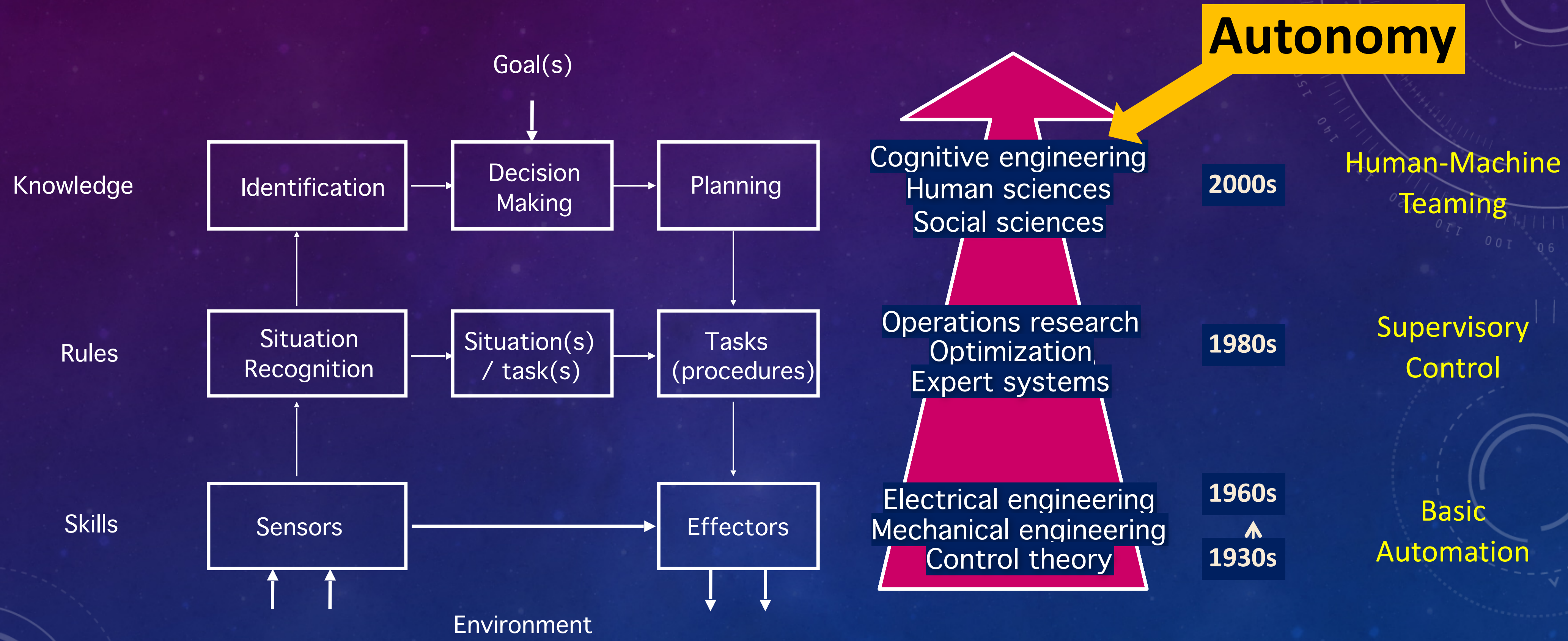
HCD of sociotechnical systems in a **digital world**

- Co-designing Technology, **Organization** and People's activities (TOP Model)
- Think about the **life cycle of systems**


Human-Factor & Ergonomics
Task & Activity Analysis
Human & Organizational
Performance Evaluation & Metrics
Integrated Logistics Support & Maintenance
Information Technology
Human-Computer Interaction
Artificial Intelligence
Visualization techniques
Modeling & Simulation
Workforce Planning
Safety
Environment
Competences & Professionalism
Human Factors Engineering
People
Organization
HSI
Sustainability
HSI Planning
Technology
Habitability
Training
Systems Engineering
Sociotechnical Systems
Systems of Systems
Agile Development
Design & System thinking
Model-Based SE
Social, Cultural & Organizational Factors
UX/Usability
Occupational Health
Operational Domain
Operational Expertise & Experience
Scenario Elicitation
Experimental test personnel

FROM AUTOMATION TO AUTONOMY…

… and emergence of contributing disciplines (Rasmussen's model)



Boy, G.A. & Morel, C. (2022 to appear). The Machine as a Partner: Human-Machine Teaming Design using the PRODEC Method. WORK Journal.

TASK VS. ACTIVITY

1950s	1960s	1970s	1980s	1990s	2000s	2010s

Observe **activity** before design & when everything is built

HFE

(corrective ergonomics)

Analyze and take into account **tasks** at design time

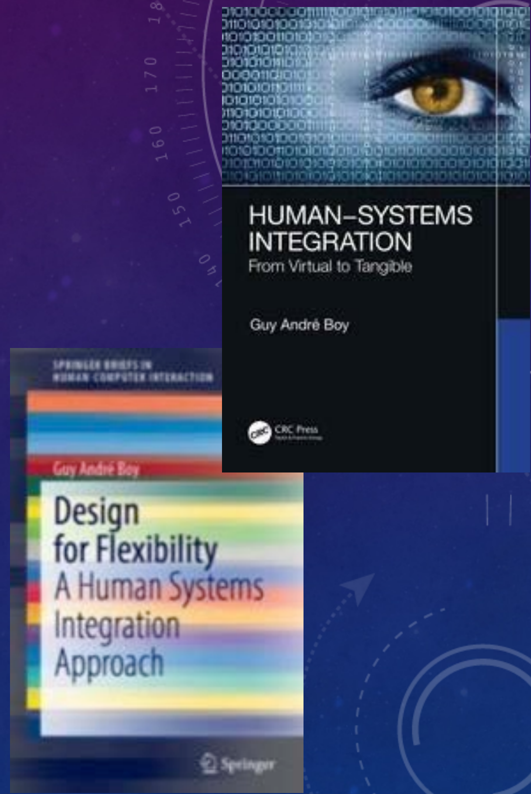
HCI

(interaction design)

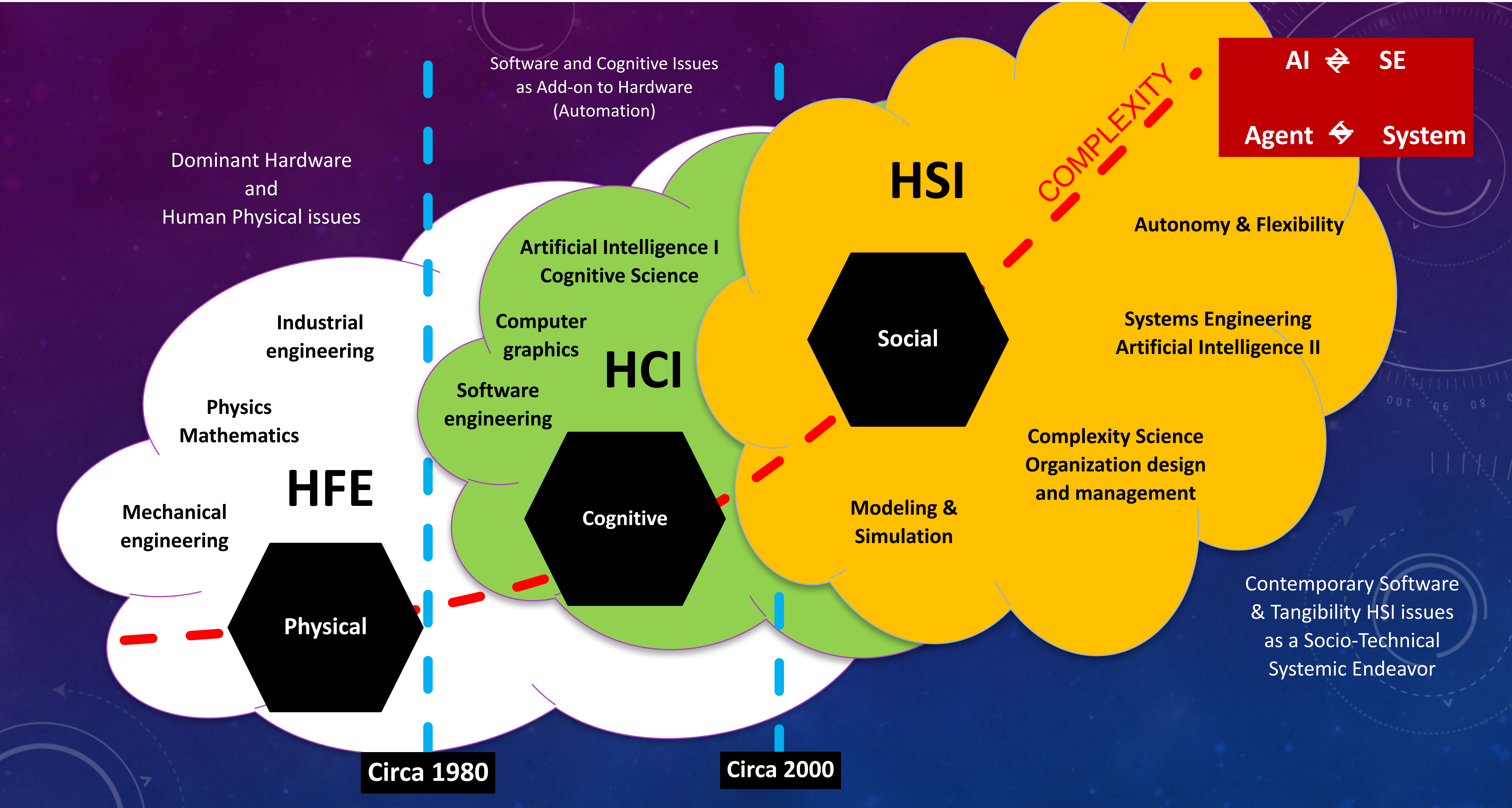
Observe and take into account **activity** at design time

HSI

(Tangible VHCD in SE)

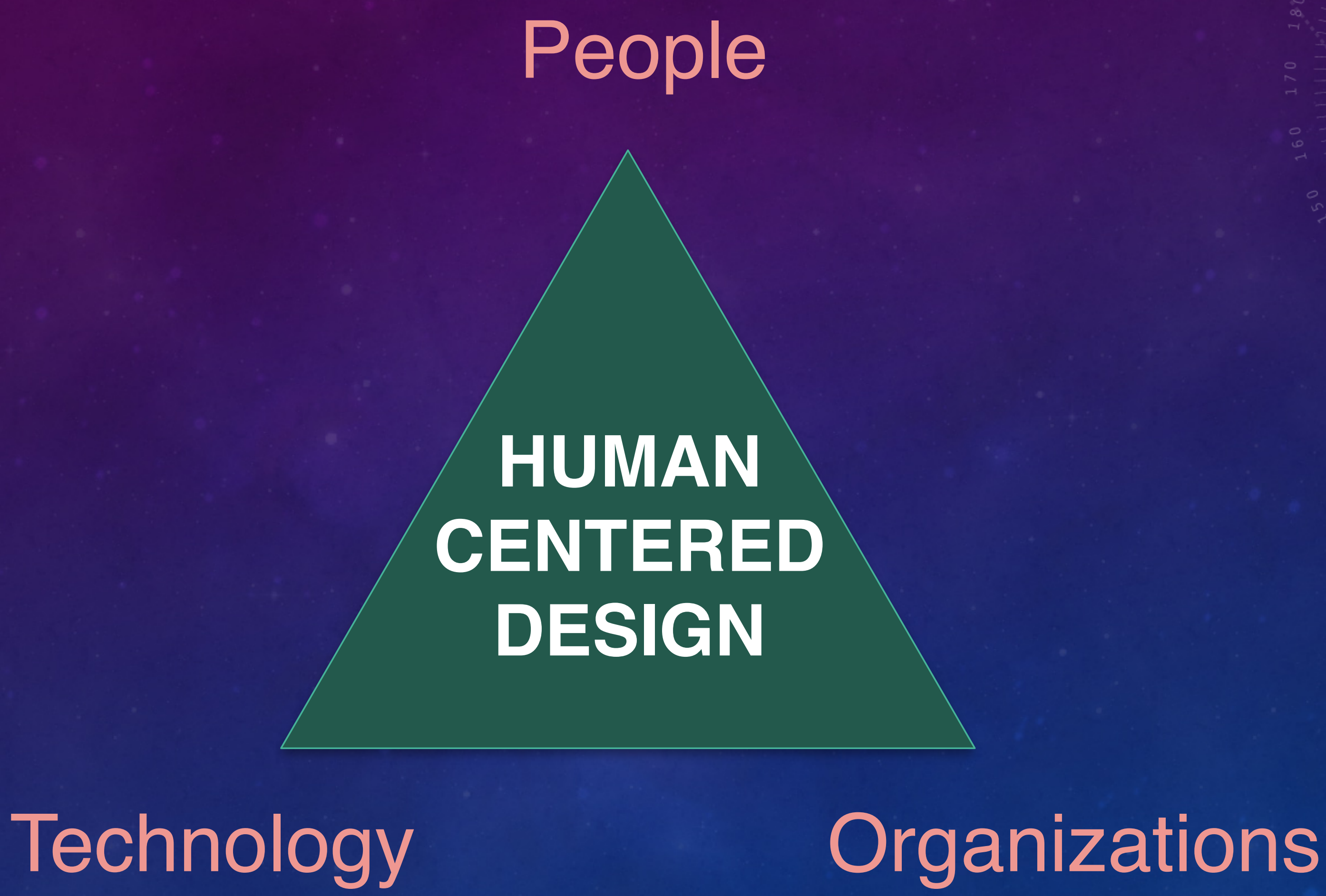


HFE: Human Factors and Ergonomics
HCI: Human Computer Interaction
VHCD: Virtual Human-Centered Design
HSI: Human Systems Integration
SE: Systems Engineering


Software and Cognitive Issues
as Add-on to Hardware
(Automation)
Dominant Hardware
and
Human Physical issues
AI ⇔ SE
Agent ⇔ System
COMPLEXITY
HSI
Autonomy & Flexibility
Artificial Intelligence I
Cognitive Science
Computer
graphics
Software
engineering
HCI
Industrial
engineering
Physics
Mathematics
HFE
Mechanical
engineering
Social
Systems Engineering
Artificial Intelligence II
Complexity Science
Organization design
and management
Modeling &
Simulation
Cognitive
Physical
Contemporary Software
& Tangibility HSI issues
as a Socio-Technical
Systemic Endeavor
Circa 1980
Circa 2000

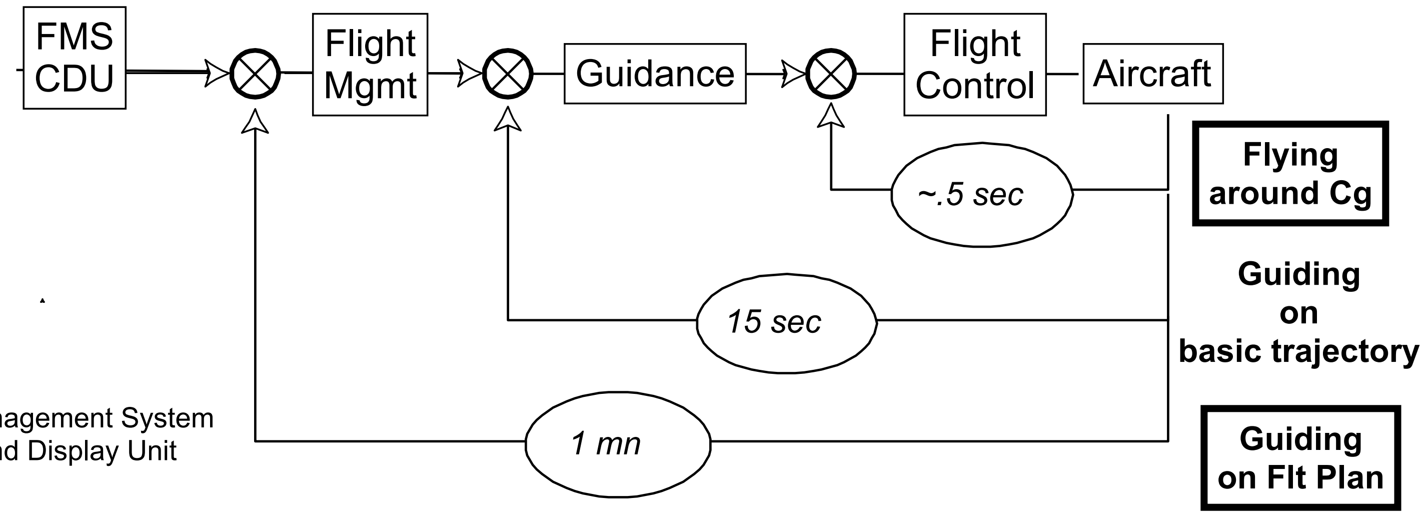
COMPLICATED	COMPLEX
Make it simpler, more rational, and usable	Make it familiar, understandable, and useful

THE COMPLICATED-COMPLEX DISTINCTION


People
HUMAN
CENTERED
DESIGN
Technology
Organizations

Loop 3: Navigation automation

Glass cockpits



FMS: Flight Management System
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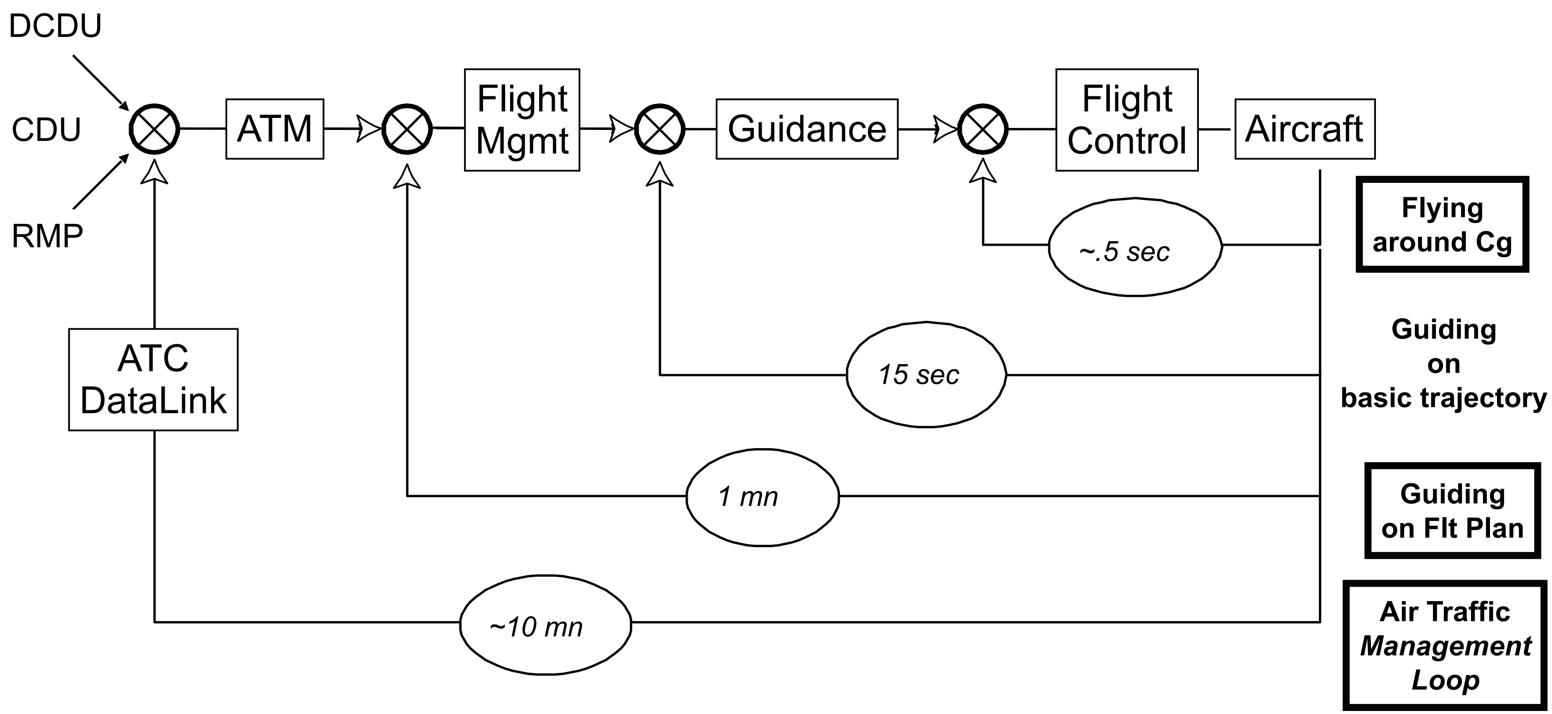
Integration of guidance and flight management

From control to management (revolution)

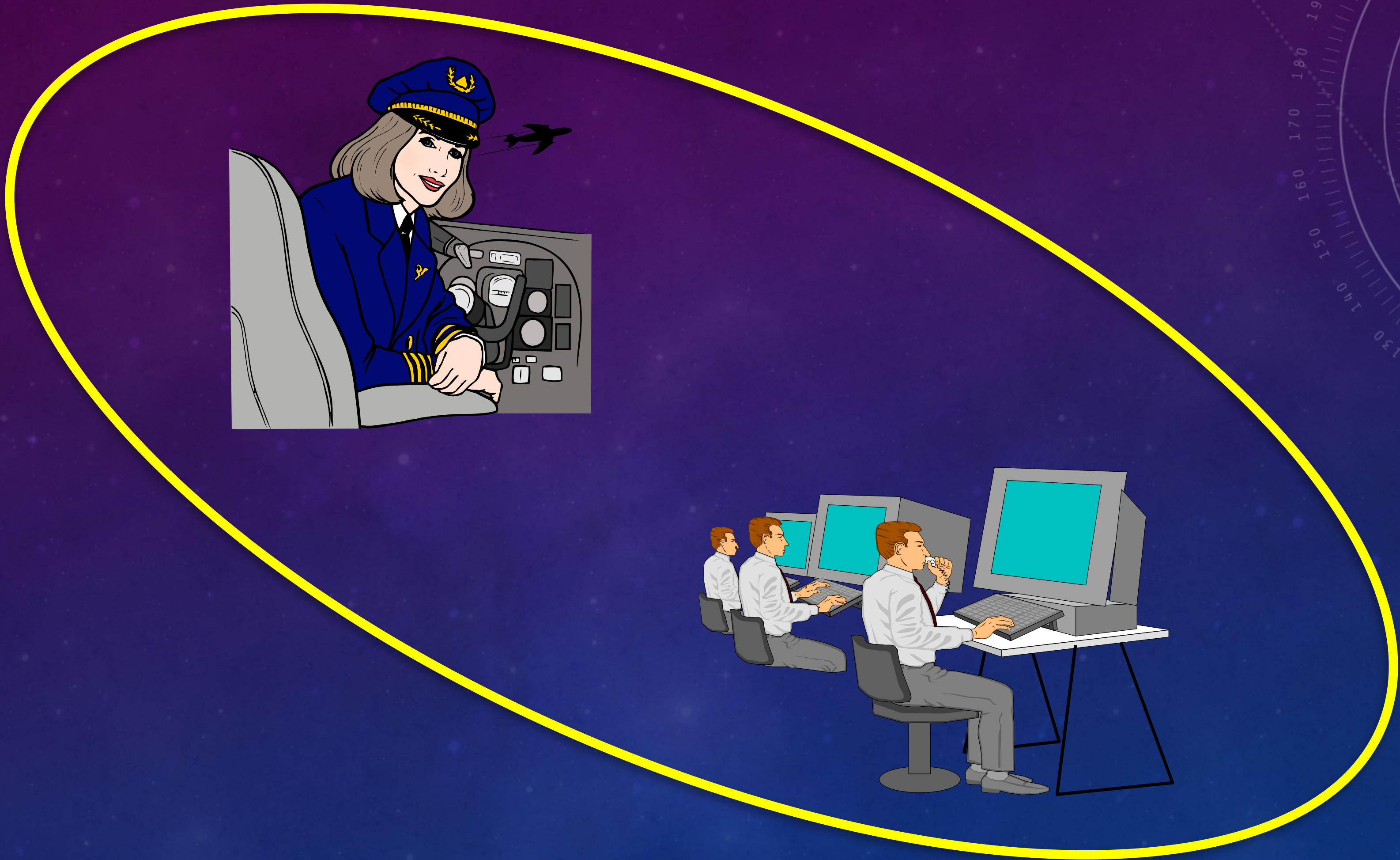
The 4-loop approach to ATM was first provided by Etienne Tarnowski at HCI-Aero'06, Seattle, USA

Loop 4: Automation of the ATM



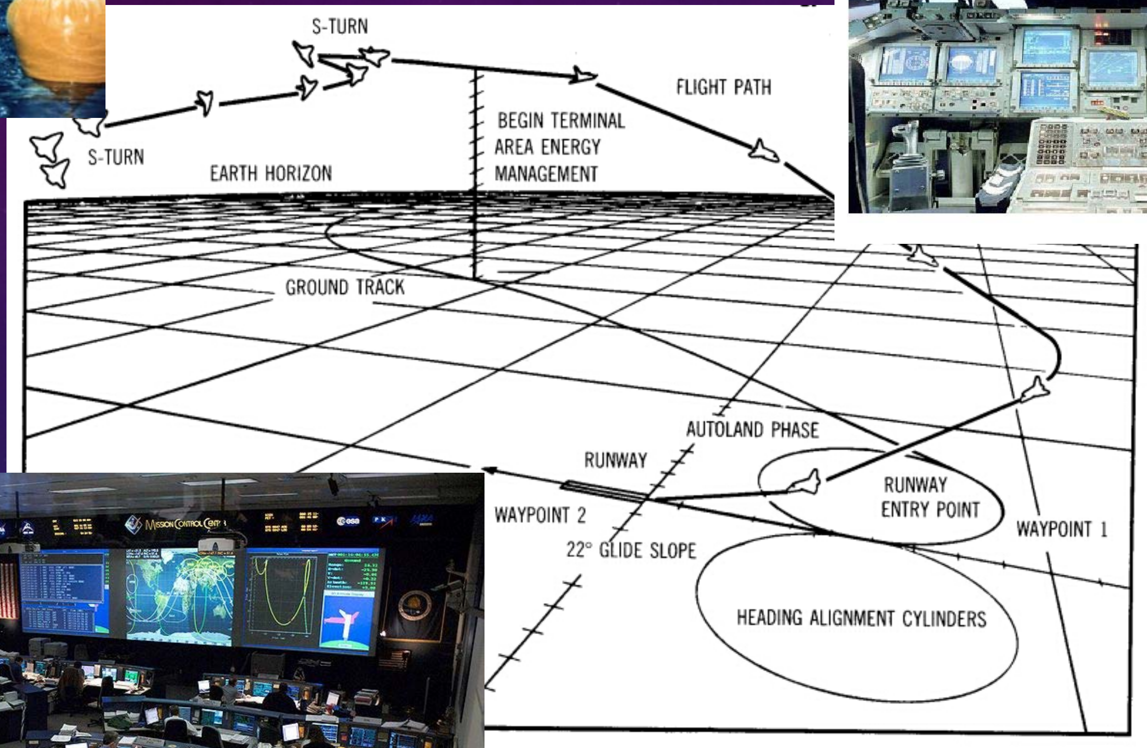
DCDU: Datalink Control & Display Unit
CDU: Control and Display Unit
RMP: Radio Management Panel

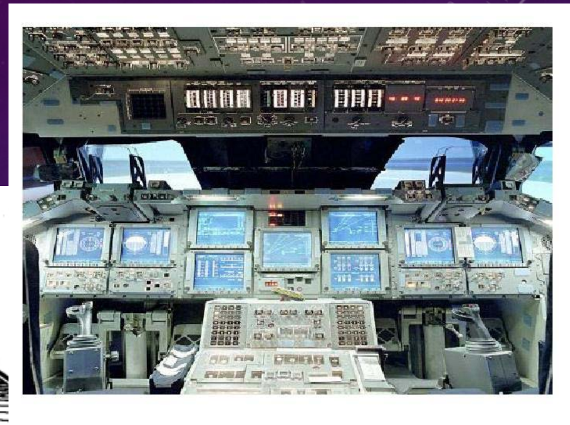
The 4-loop approach to ATM was first provided by Etienne Tarnowski at HCI-Aero'06, Seattle, USA



SPACE SHUTTLE COCKPIT

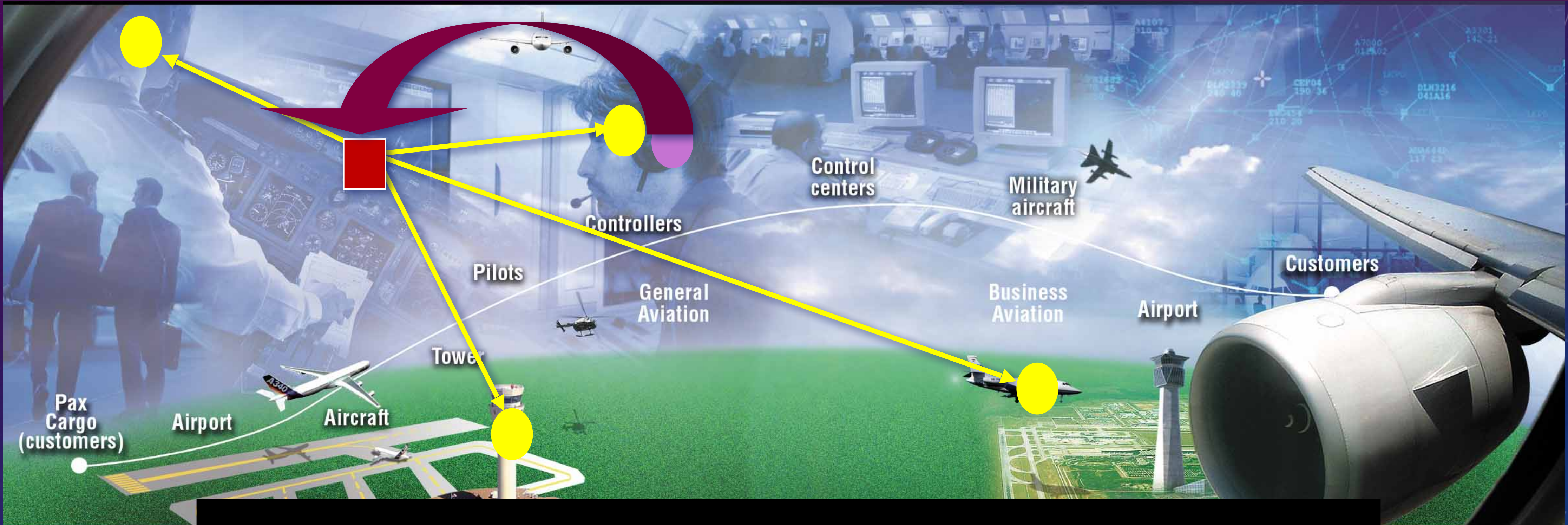
MISSION CONTROL CENTER


S-TURN
FLIGHT PATH
S-TURN
EARTH HORIZON
BEGIN TERMINAL
AREA ENERGY
MANAGEMENT
GROUND TRACK
AUTOLAND PHASE
RUNWAY
RUNWAY
ENTRY POINT
WAYPOINT 2
WAYPOINT 1
22° GLIDE SLOPE
HEADING ALIGNMENT CYLINDERS



Integration requires a **model** that represents **agents** involved and **relationships** within the **overall system**

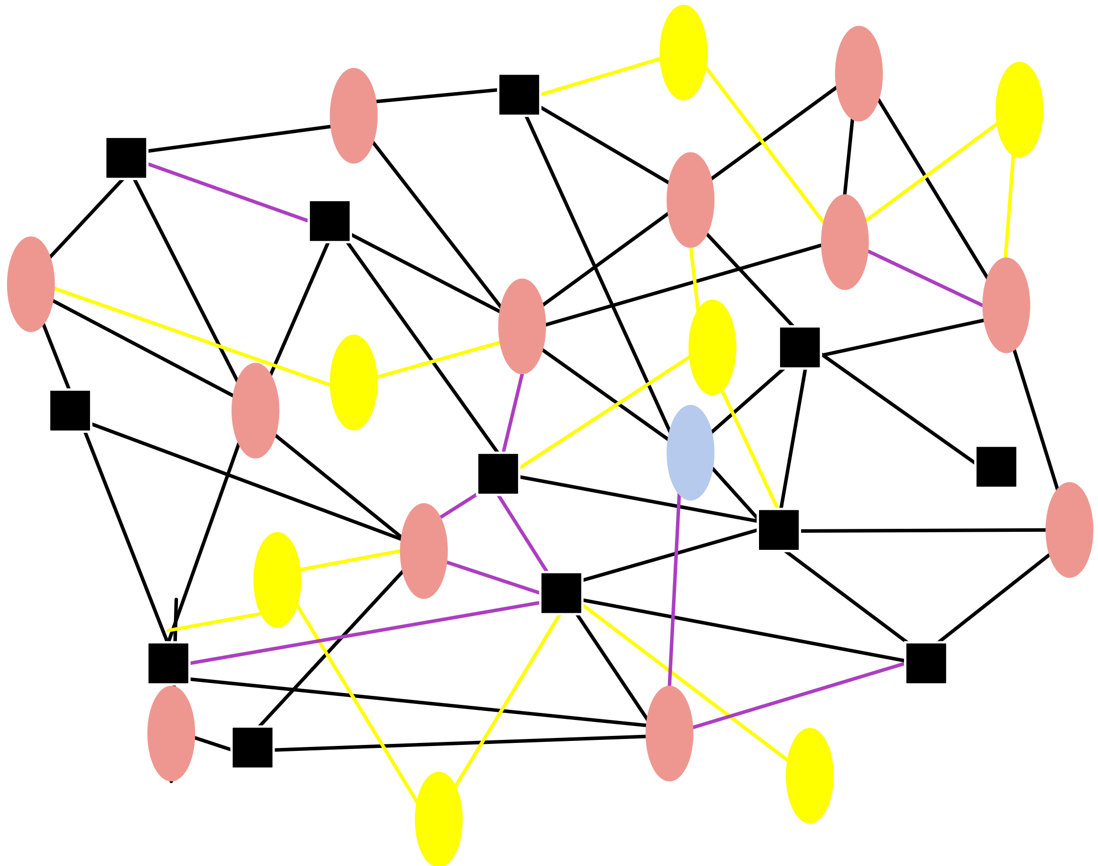
i.e., an **Organizational** Model



Machine cognitive function

Human cognitive function

A MULTI-AGENT SYSTEM…



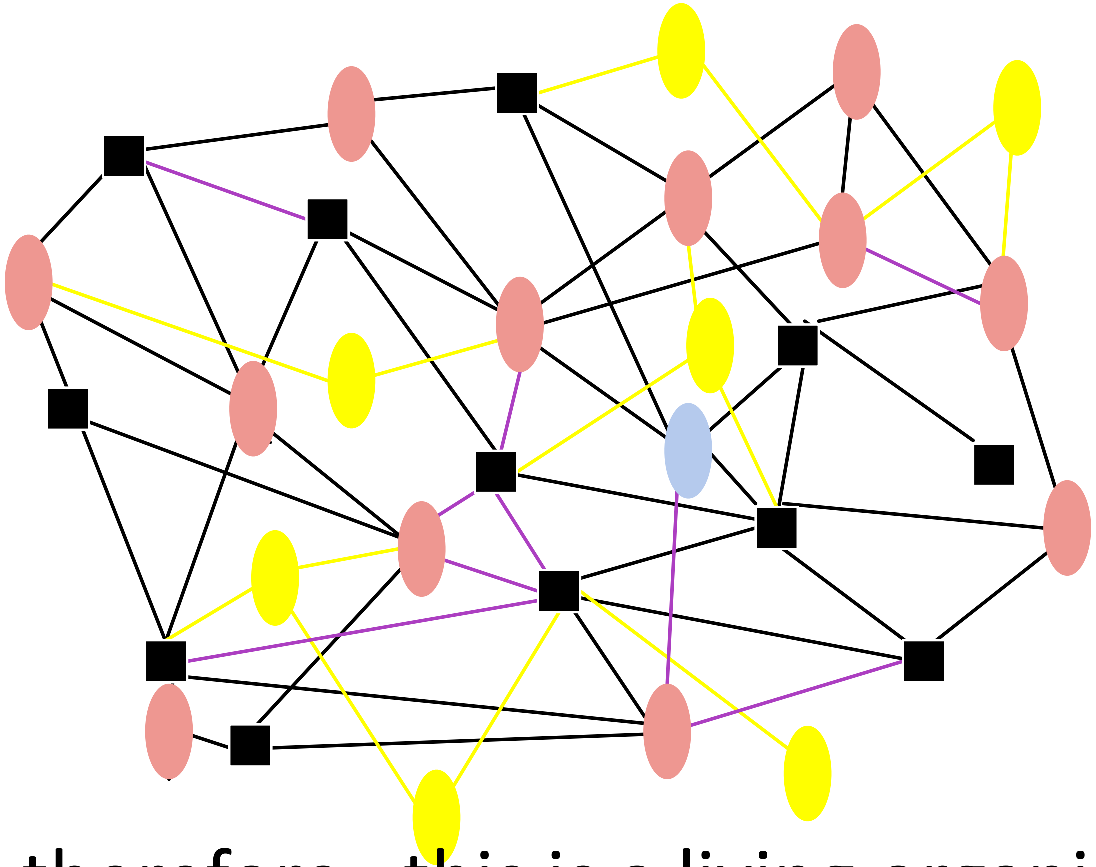
a living organism

MULTI-AGENT SYSTEMS PROPERTIES…

Separability
a crucial issue

Complexity
in connections
as well as
in agents
themselves

Emergent functions
and
the **maturity** issue



… therefore, this is a living organism

THE ORCHESTRA MODEL

http://www.amazon.com/Orchestrating-Human-Centered-Design-Guy-Boy/dp/1447143388

Air Traffic Management…

Flying in the early 21st century, in **high density traffic**, requires new competencies and systems capable of handling **complexity** of the overall organization.

→ **Complexity Science**

↓ Discover, model and use

Emergent Properties and Behaviors

THEREFORE, THE PROBLEM IS …

Aircraft cannot be considered as independent…
… but should be in their multi-agent environment
and taking into account ATM complexity

➡ e.g., self-separation
human & systems orchestration
looking for models of interaction

Air Traffic Management…

Air Show vs. Flock of birds

Automation

Manual & Automatic Control

Autonomy

Protection Envelopes

Tangible Interaction

GOOD NEWS …

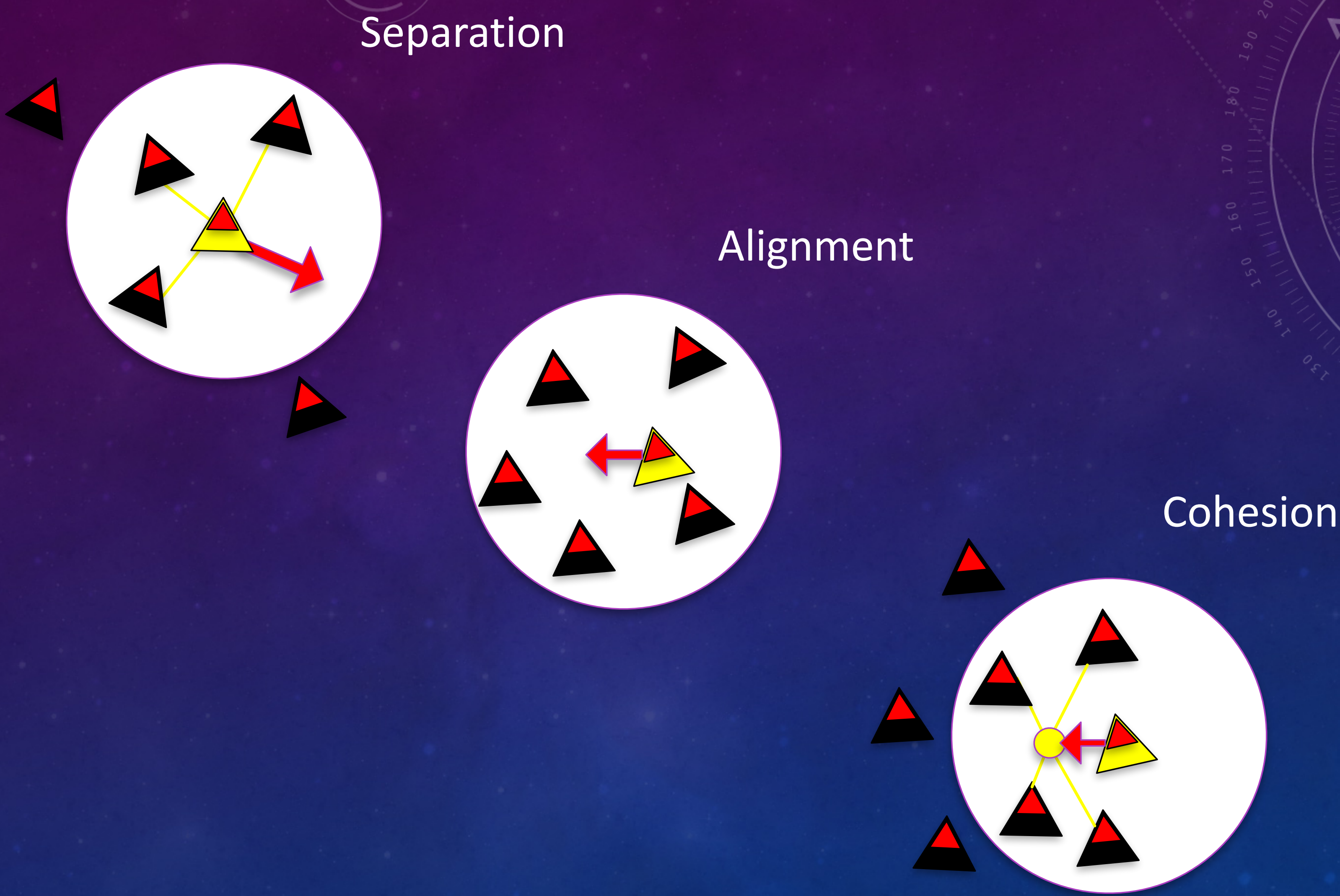
Modeling and Simulation…

… to detect emerging patterns
from multi-agent complexity

biology-inspired models
cognitive modeling
artificial intelligence

HCDi
Human-Centered Design institute


HCDi
Human-Centered Design institute



https://vanhunteradams.com/Pico/Animal_Movement/Boids-algorithm.html

WHAT IS AT STAKE?

- Automation vs. Autonomy
  - ➢ **Coordination**

The more agents are autonomous
The more they require coordination

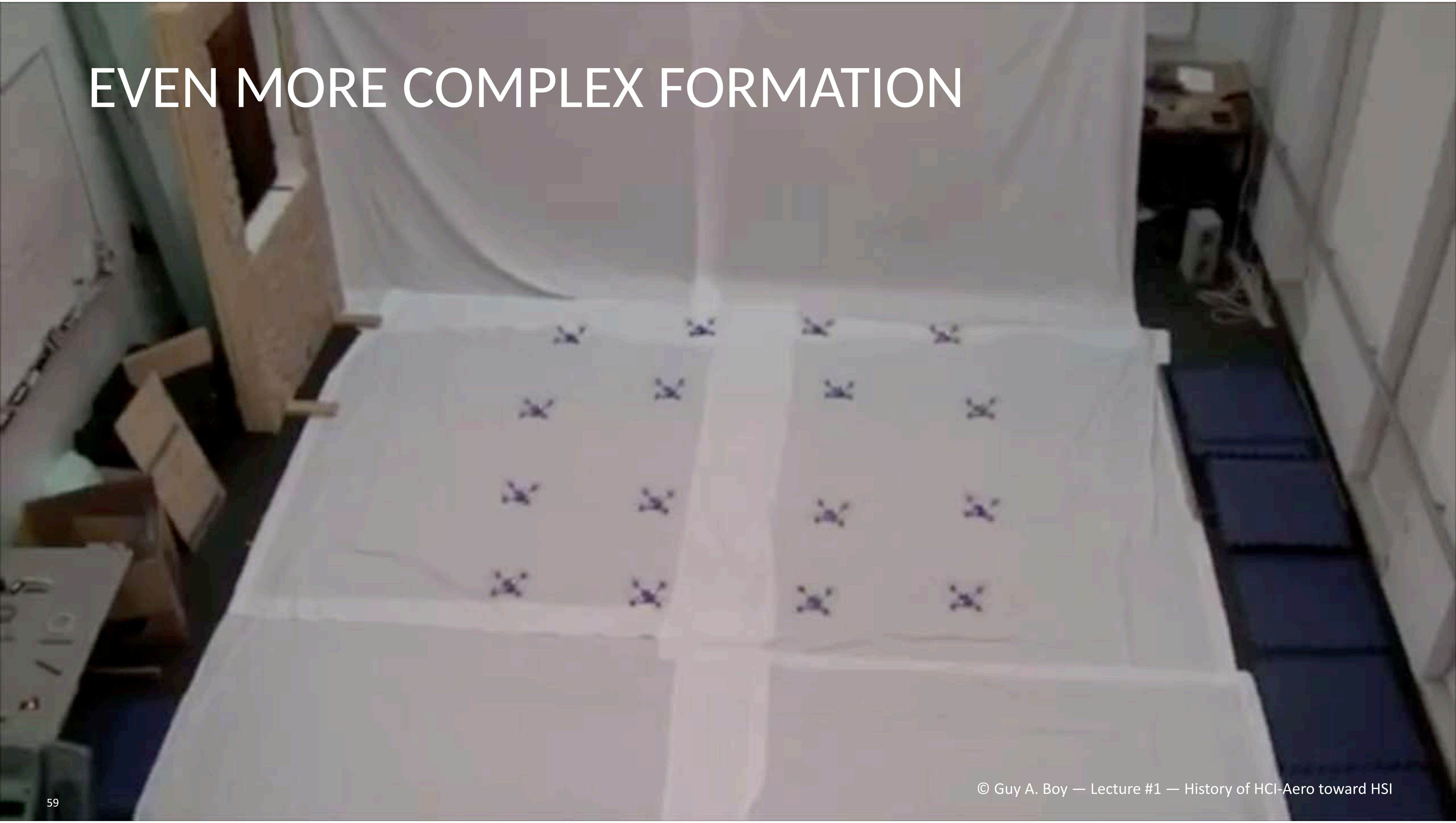
- Cognition vs. Physics
  - ➢ **Tangibility**

Something is tangible when it is **graspable**
in the **physical** sense,
but also in the **figurative** sense.

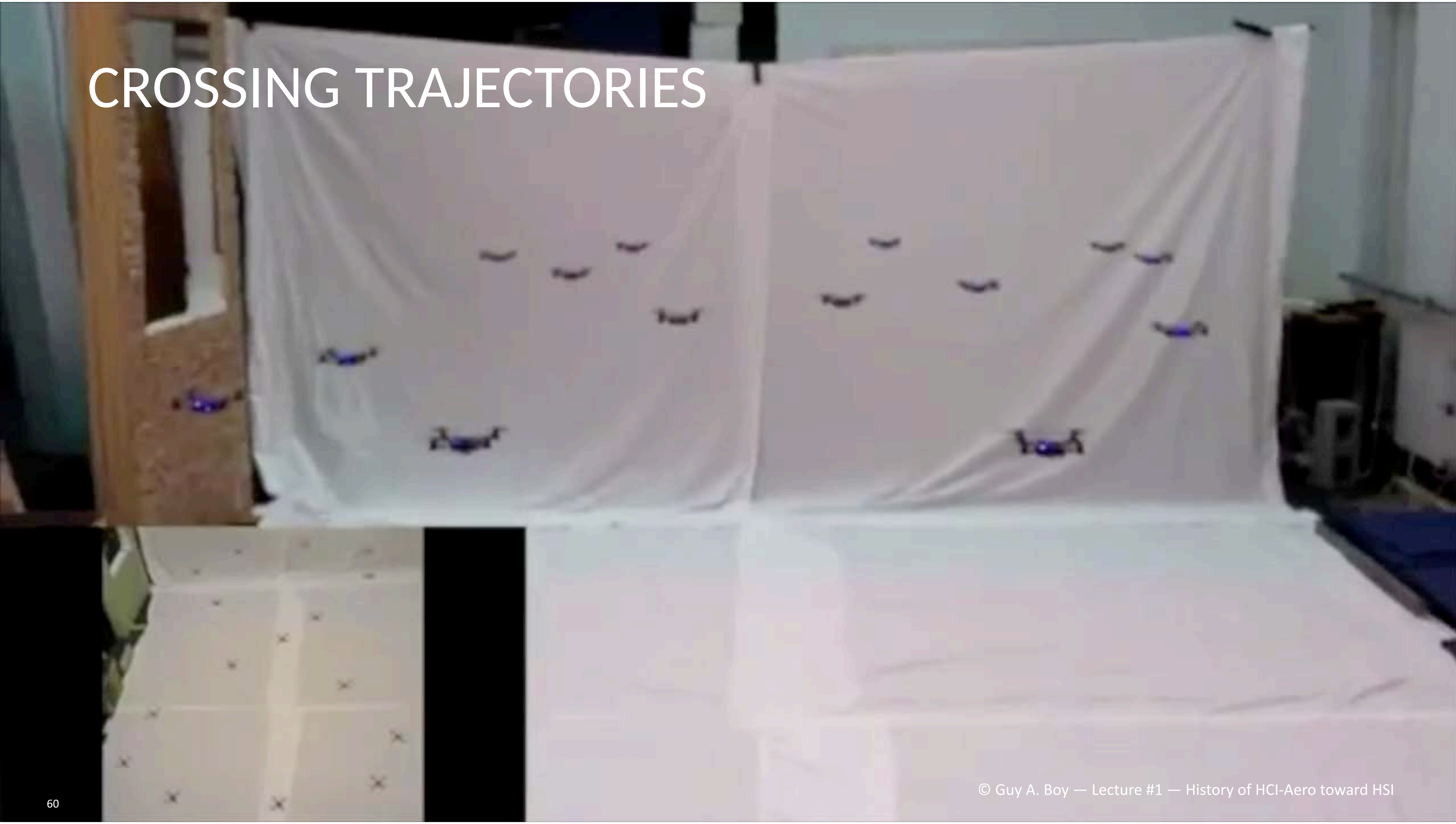
SIMPLE FORMATION

MORE COMPLEX FORMATION

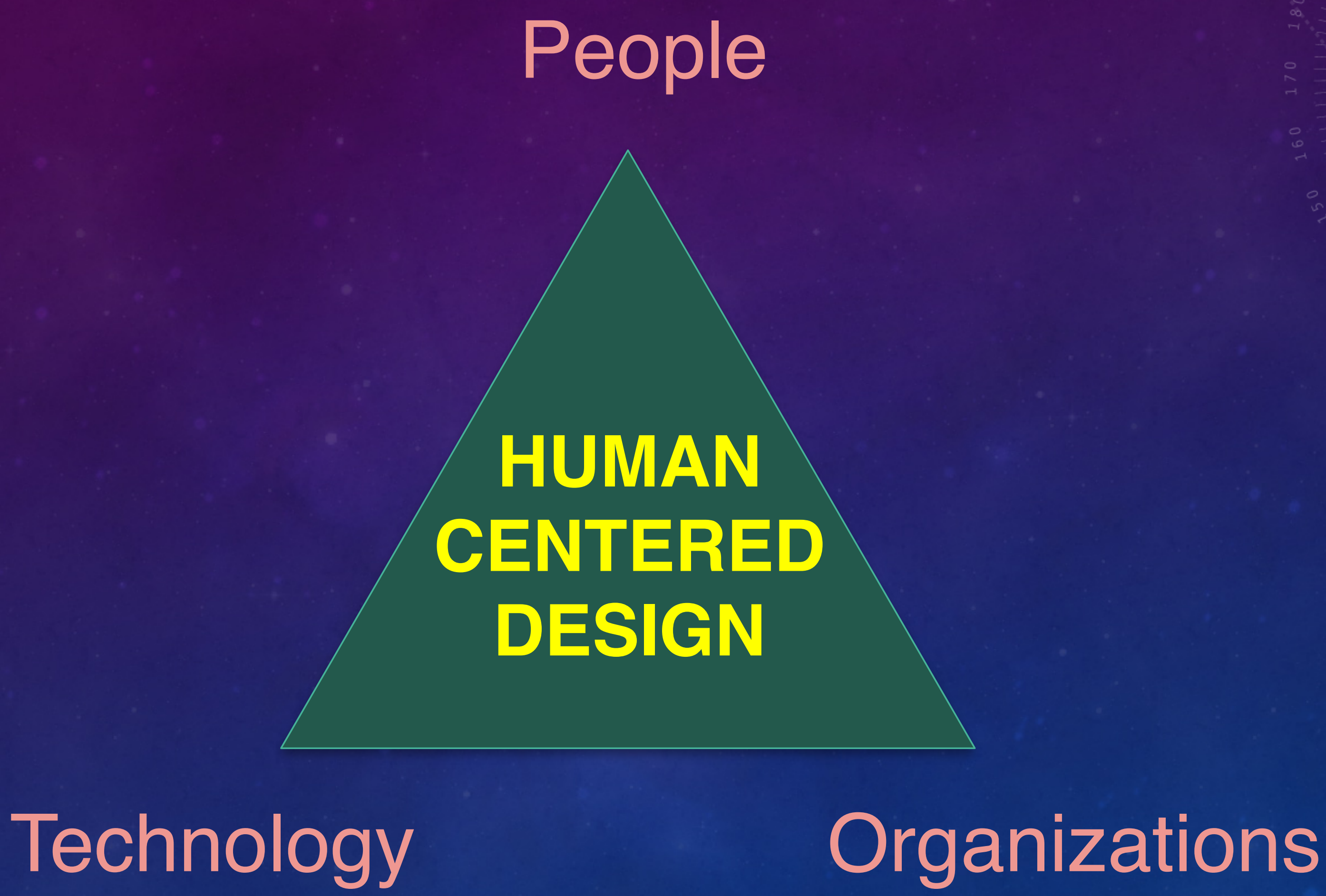
EVEN MORE COMPLEX FORMATION



CROSSING TRAJECTORIES



My Lab at Florida Institute of Technology


People
HUMAN
CENTERED
DESIGN
Technology
Organizations

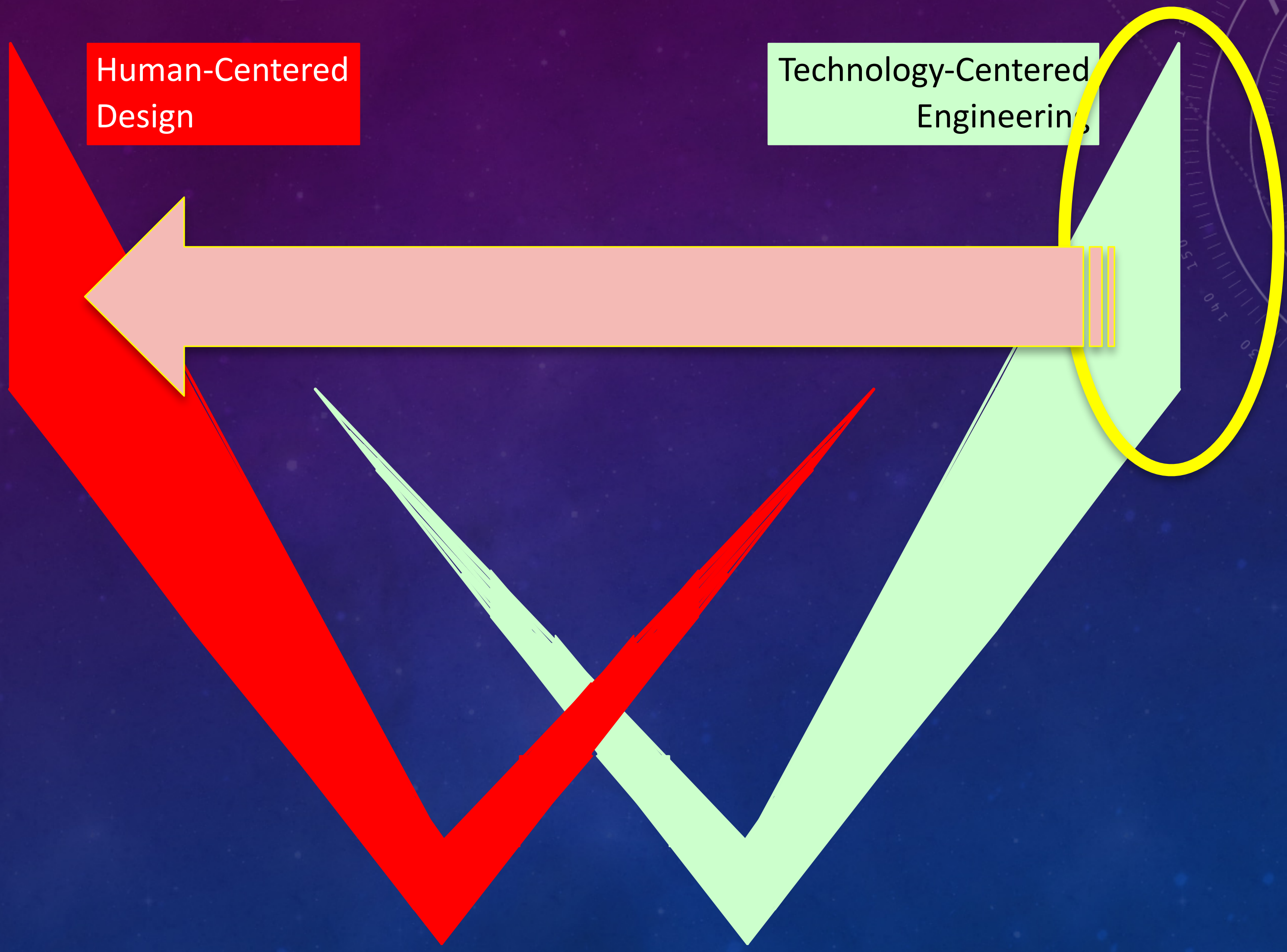
V-Model

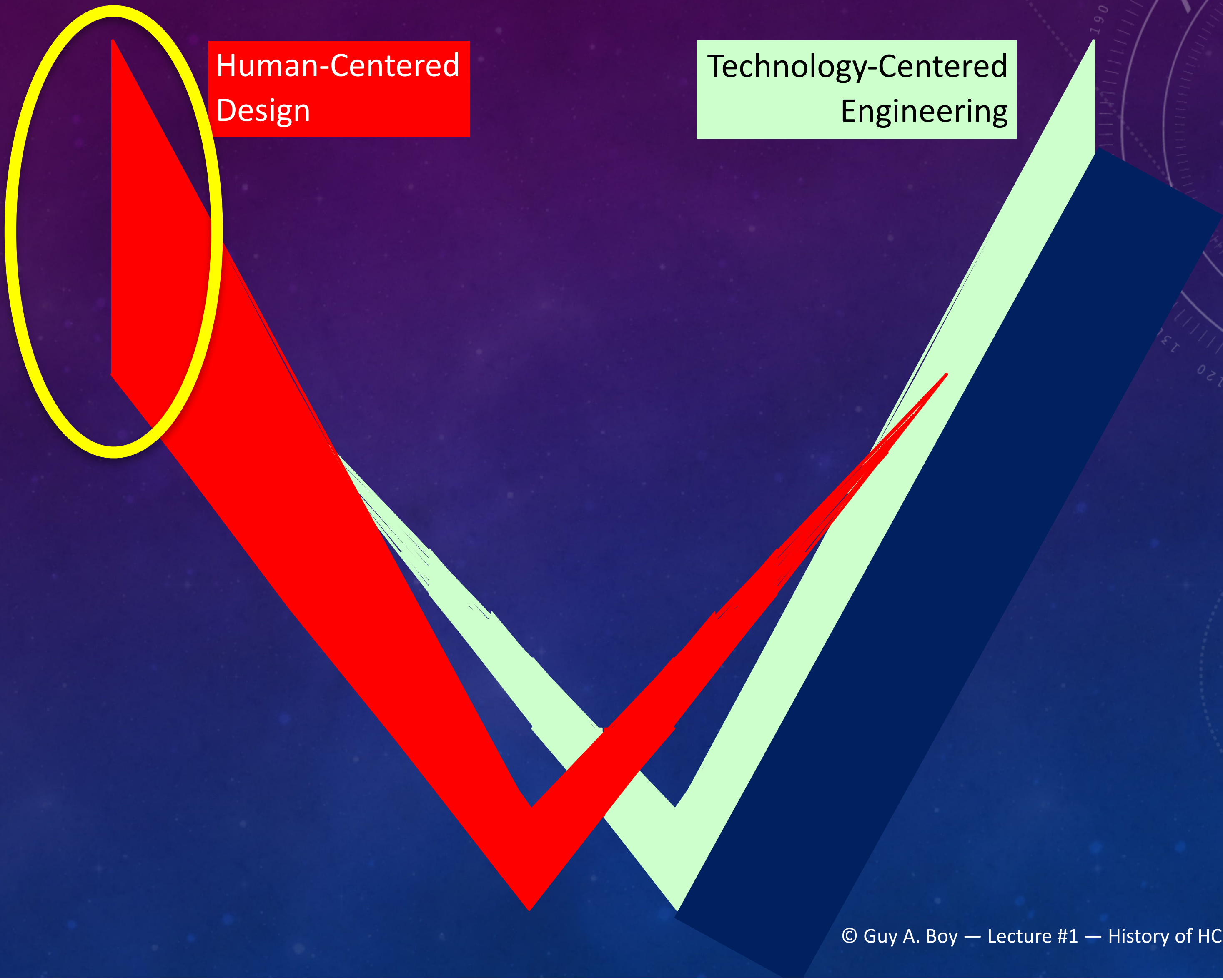
Design

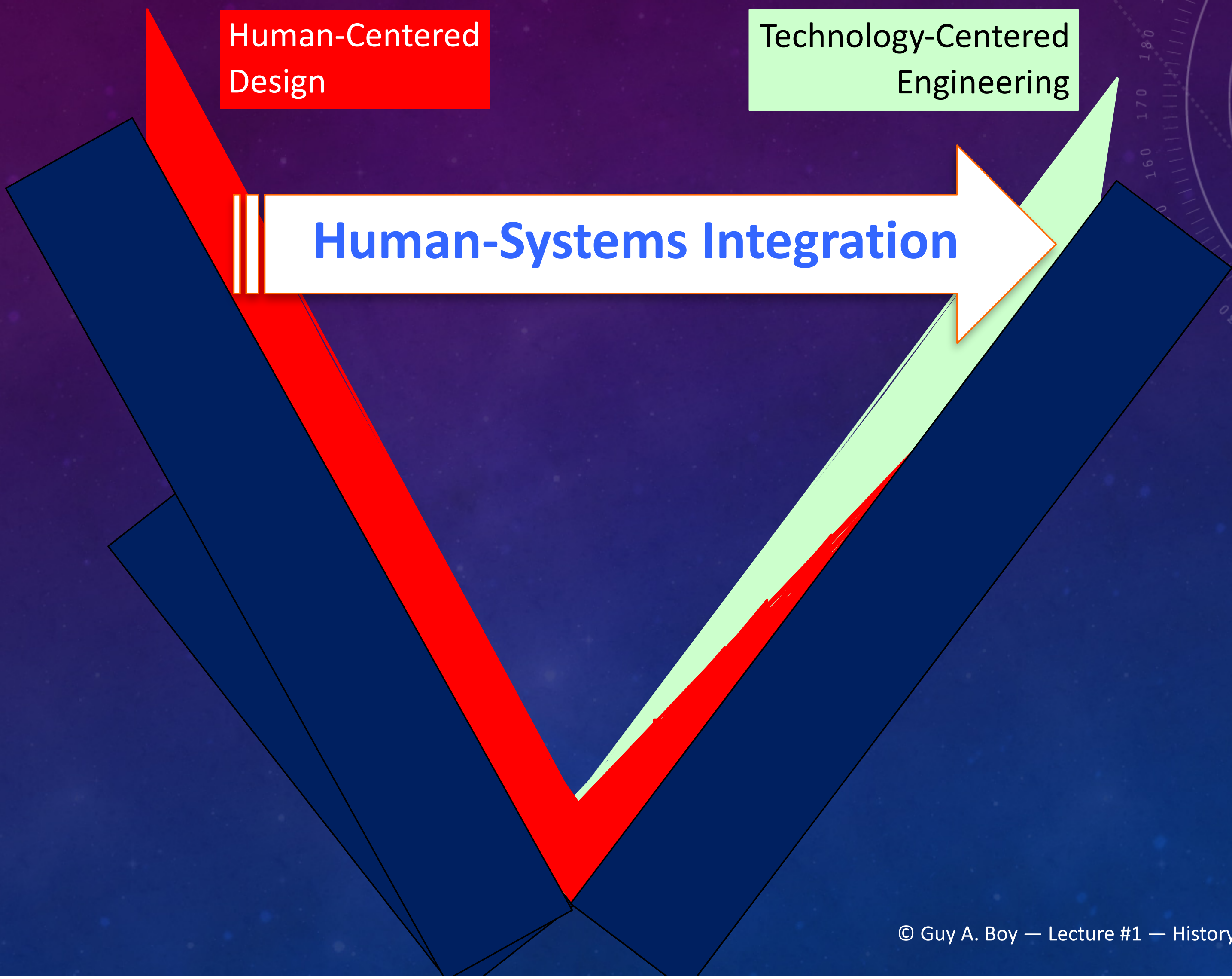
Certification

Development

Evaluation


Human-Centered Design
Technology-Centered Engineering


Human-Centered Design
Technology-Centered Engineering


Human-Centered Design
Technology-Centered Engineering
Human-Systems Integration

EXERCISE

- How could you solve the problem of airspace congestion?
- Let us take the A380 as an example…
- How would you design a big aircraft?
- What are the major parameters to be considered?`
- Why did the A380 production stop?

LINKS FOR REFERENCES AND CONNECTIONS…

- https://www.flextechchair.org/GuyAndreBoy/publications.html#
- https://en.wikipedia.org/wiki/Guy_André_Boy
- https://www.linkedin.com/in/guyboy/?originalSubdomain=fr
- https://www.youtube.com/watch?v=hJQqmw7Gslc

Thank you!…